Local Vol by SANOS — hedge experiments

March 18, 2025 · captioned results

32 figures included. All 3-DTE/3-day-labelled figures are excluded.

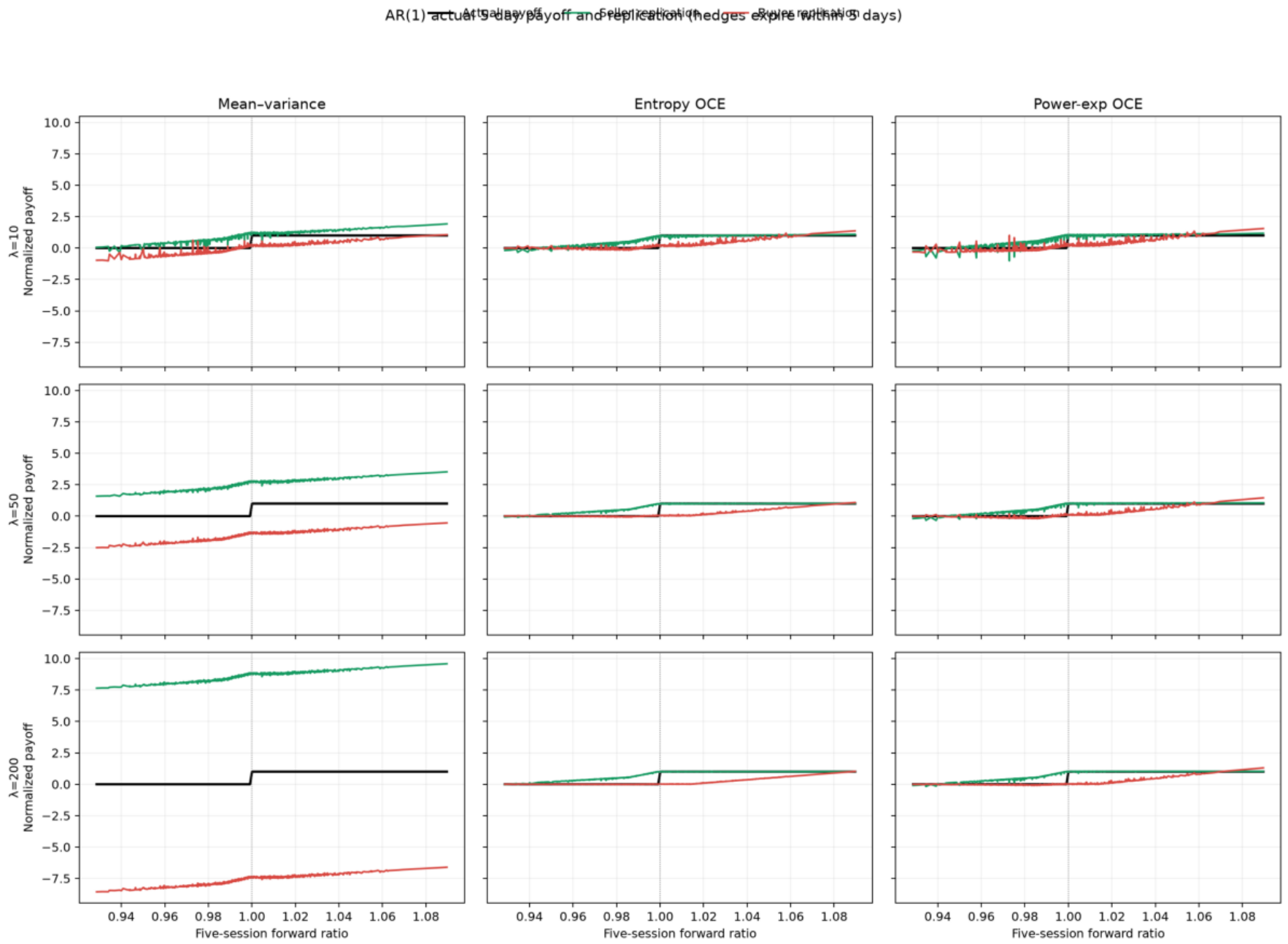
Conventions

- Target payoff: black • Seller: green • Buyer: red
- ▲ long, ▼ short • Filled strike markers: 1D; hollow: later expiry
- Size-capped runs use direct hard a_n bounds from reported volume (no tanh)
- Transaction cost is half-spread L1; regularization is the classic unscaled quadratic ridge

Corrected 5D experiment includes 1D, 2D and 5D hedge expiries and excludes only the two exact target-spread legs.

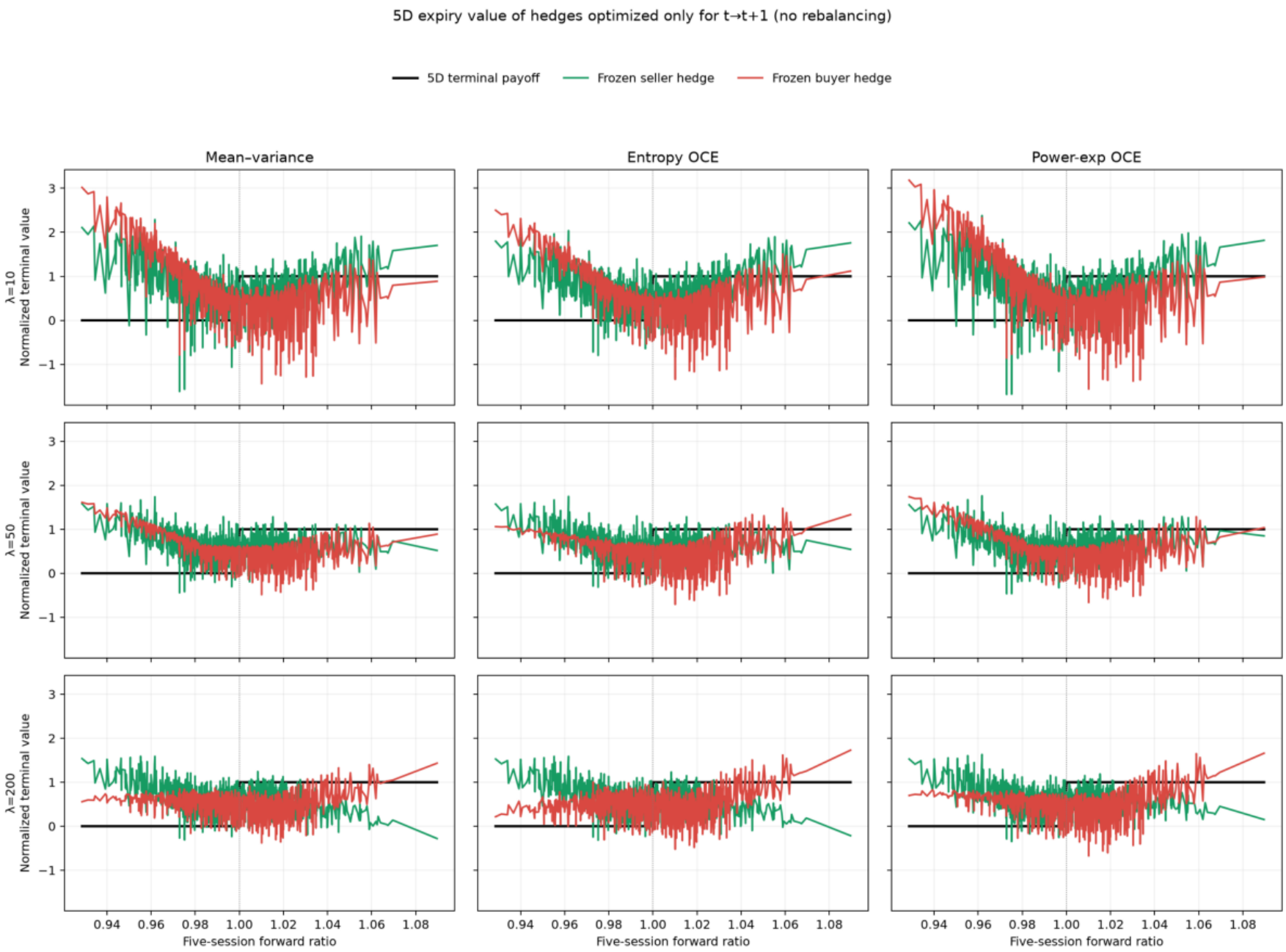
5D digital: terminal payoff replication

Actual five-session payoff in black and the seller/buyer price-plus-hedge values in green/red. Columns are utilities and rows are risk-aversion levels.



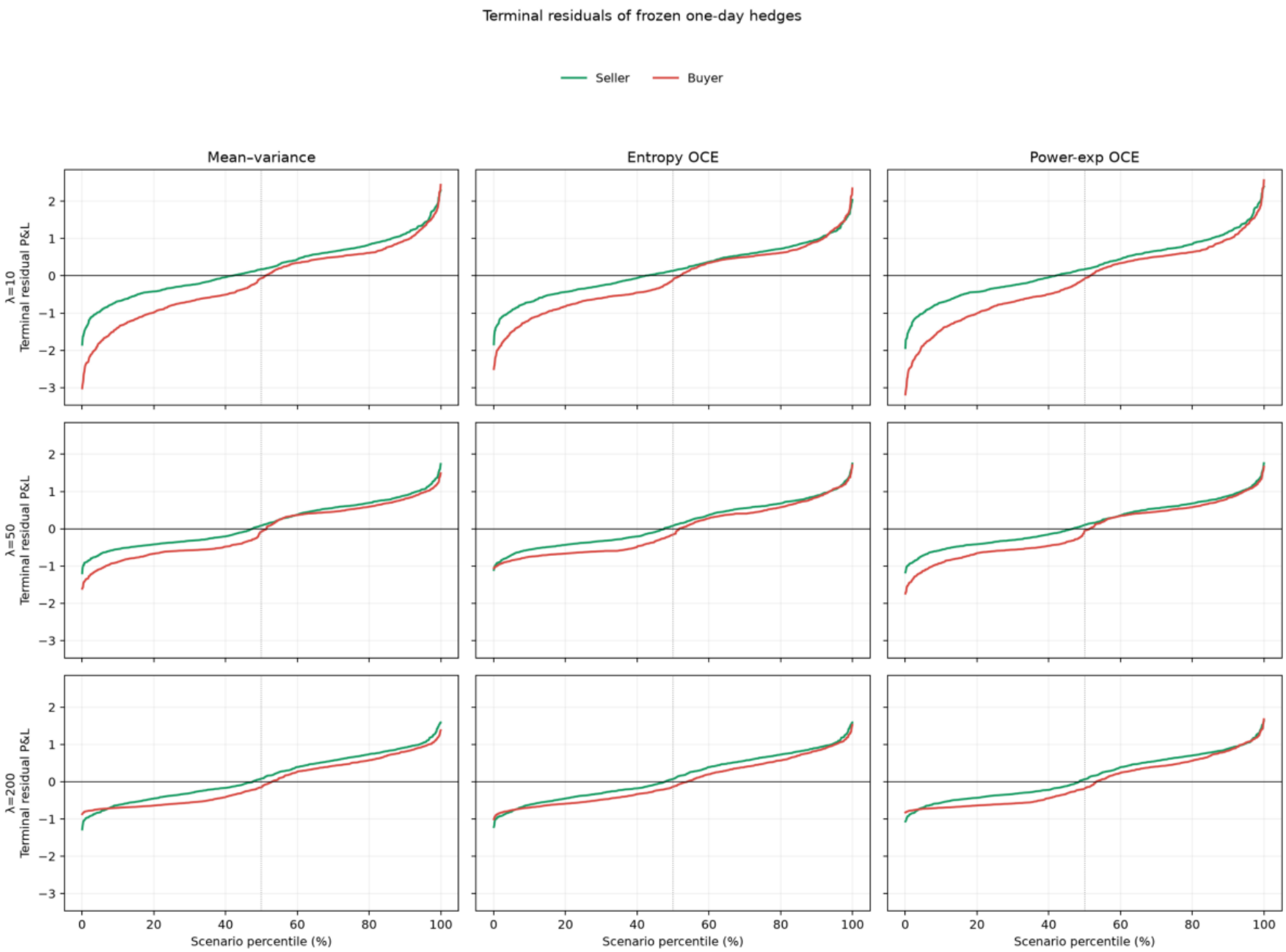
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



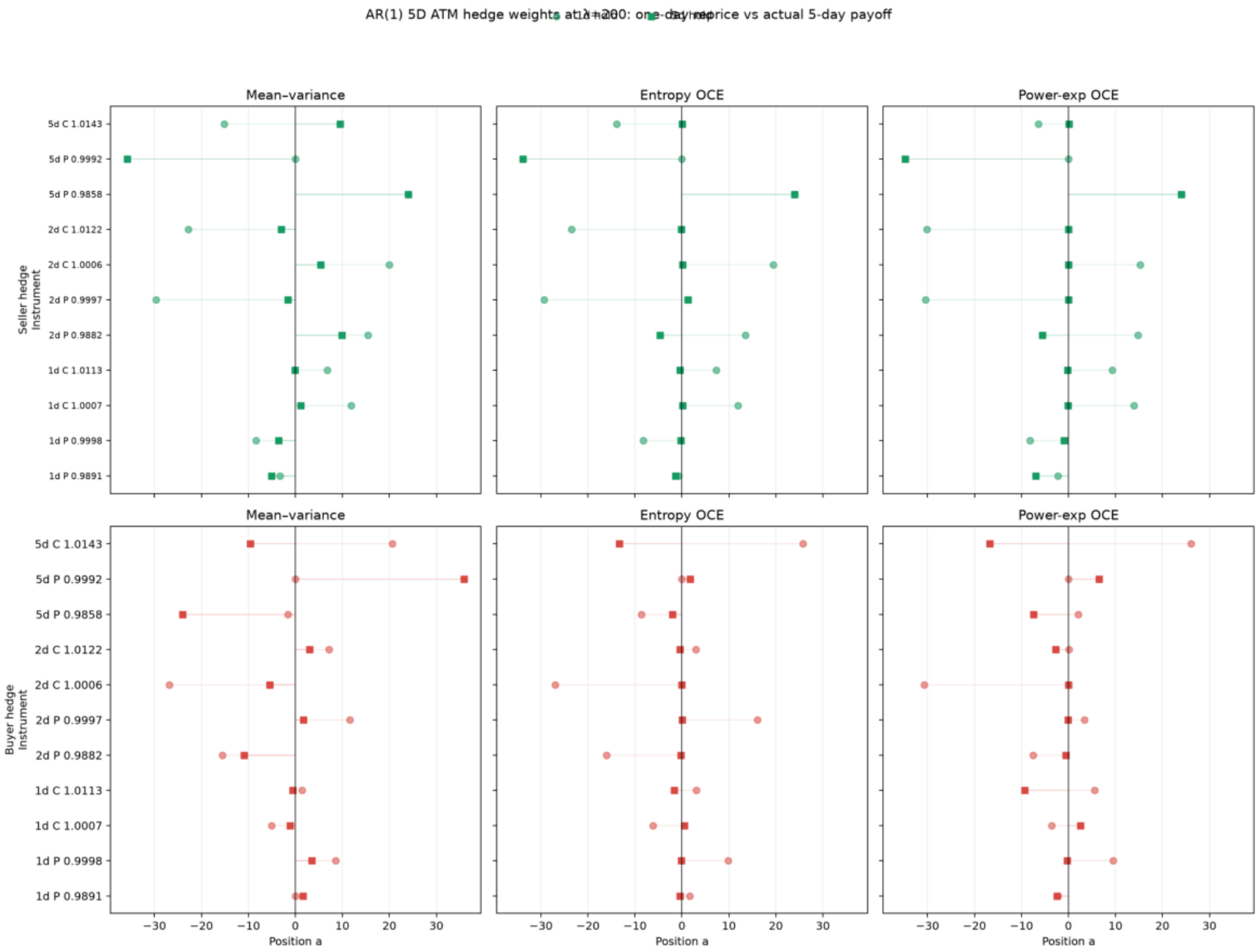
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



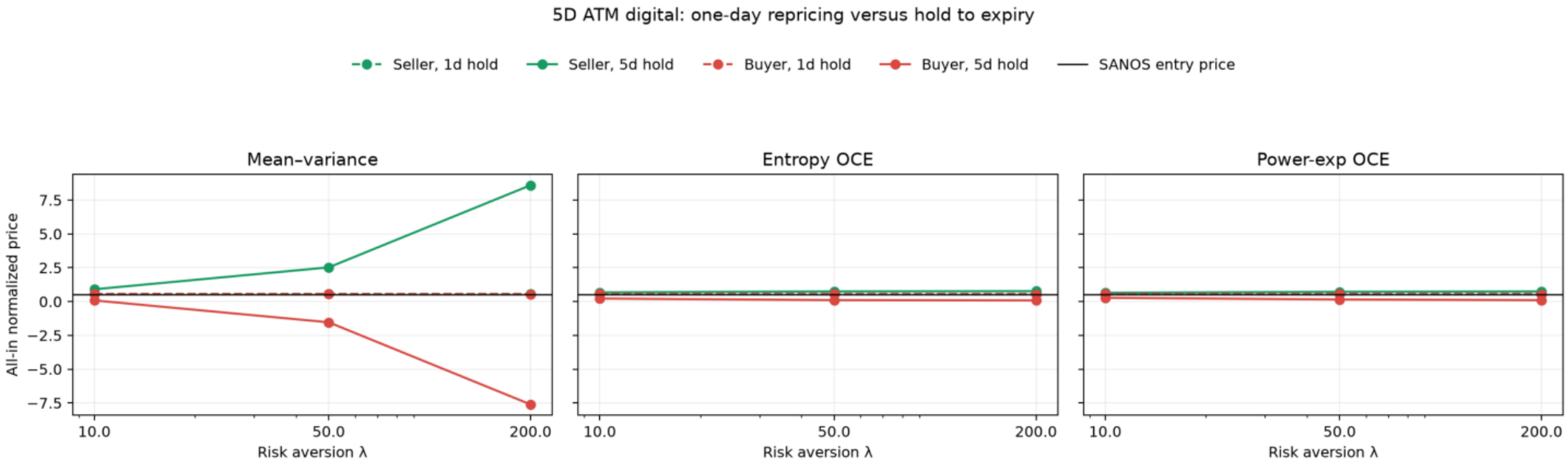
5D digital: hedge-weight comparison

At λ=200, compares buyer and seller positions for a one-day revaluation with positions held through the actual five-session payoff. The hedge universe contains 1D, 2D and 5D options.



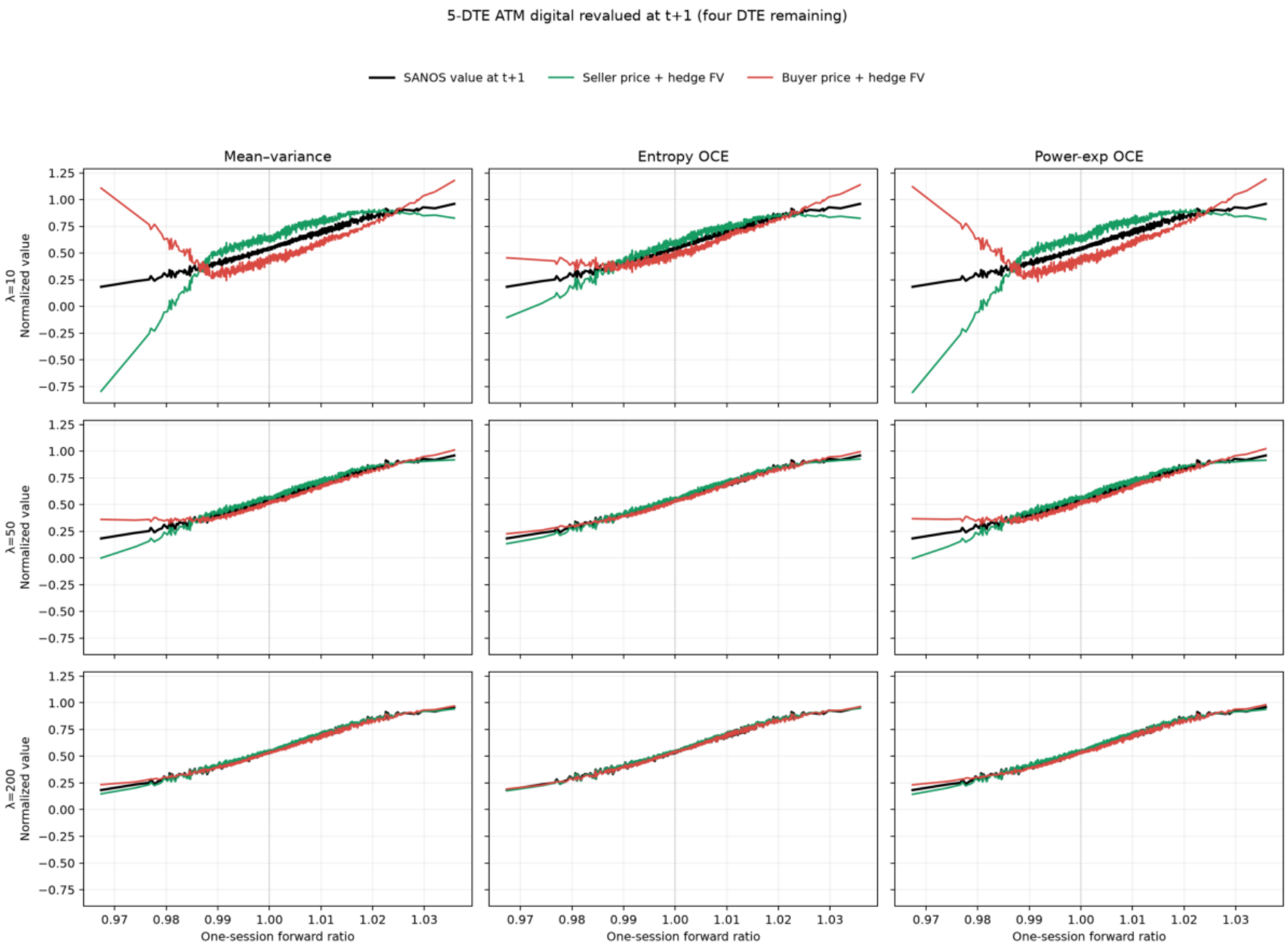
5D digital: indifference-price comparison

AR(1) seller and buyer all-in prices for a one-day SANOS revaluation versus holding the 5-DTE digital to expiry. The black line is the time-zero SANOS price.



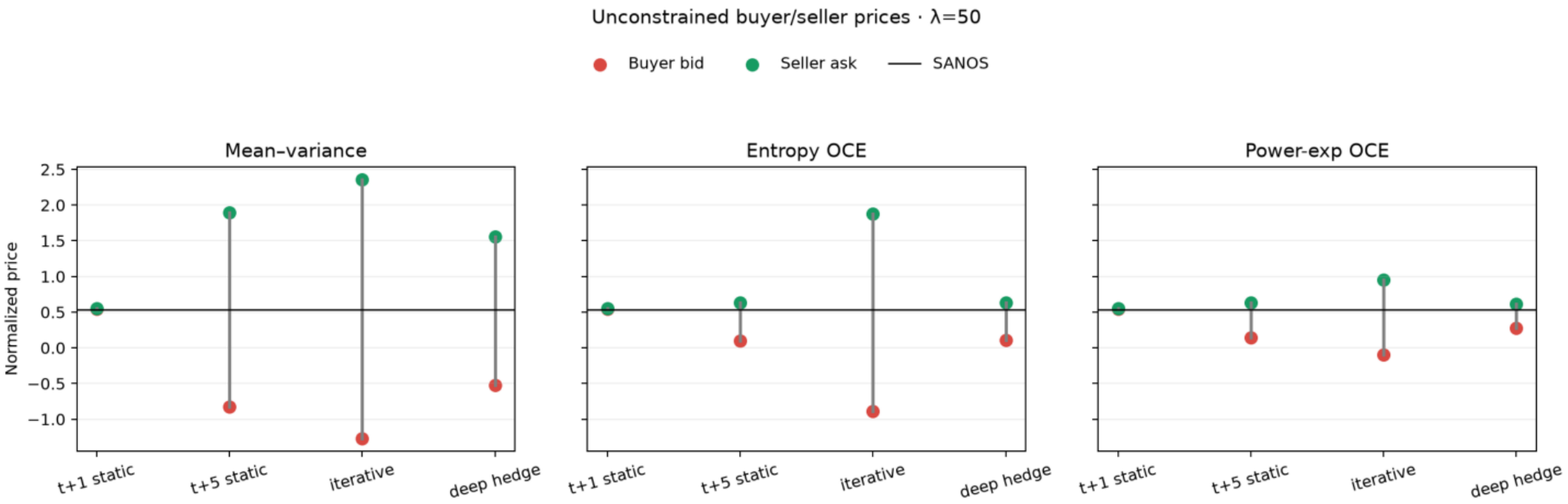
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



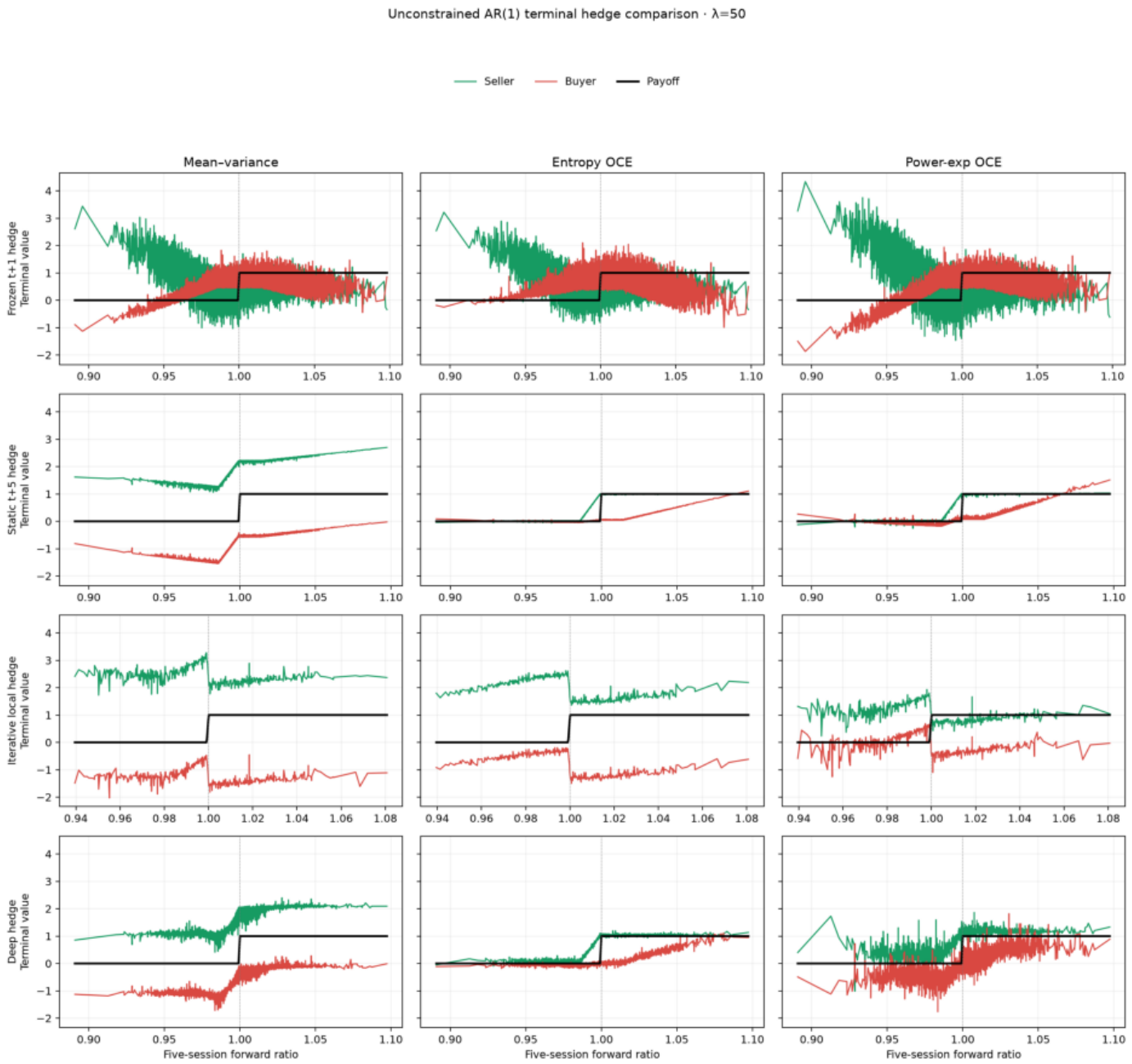
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



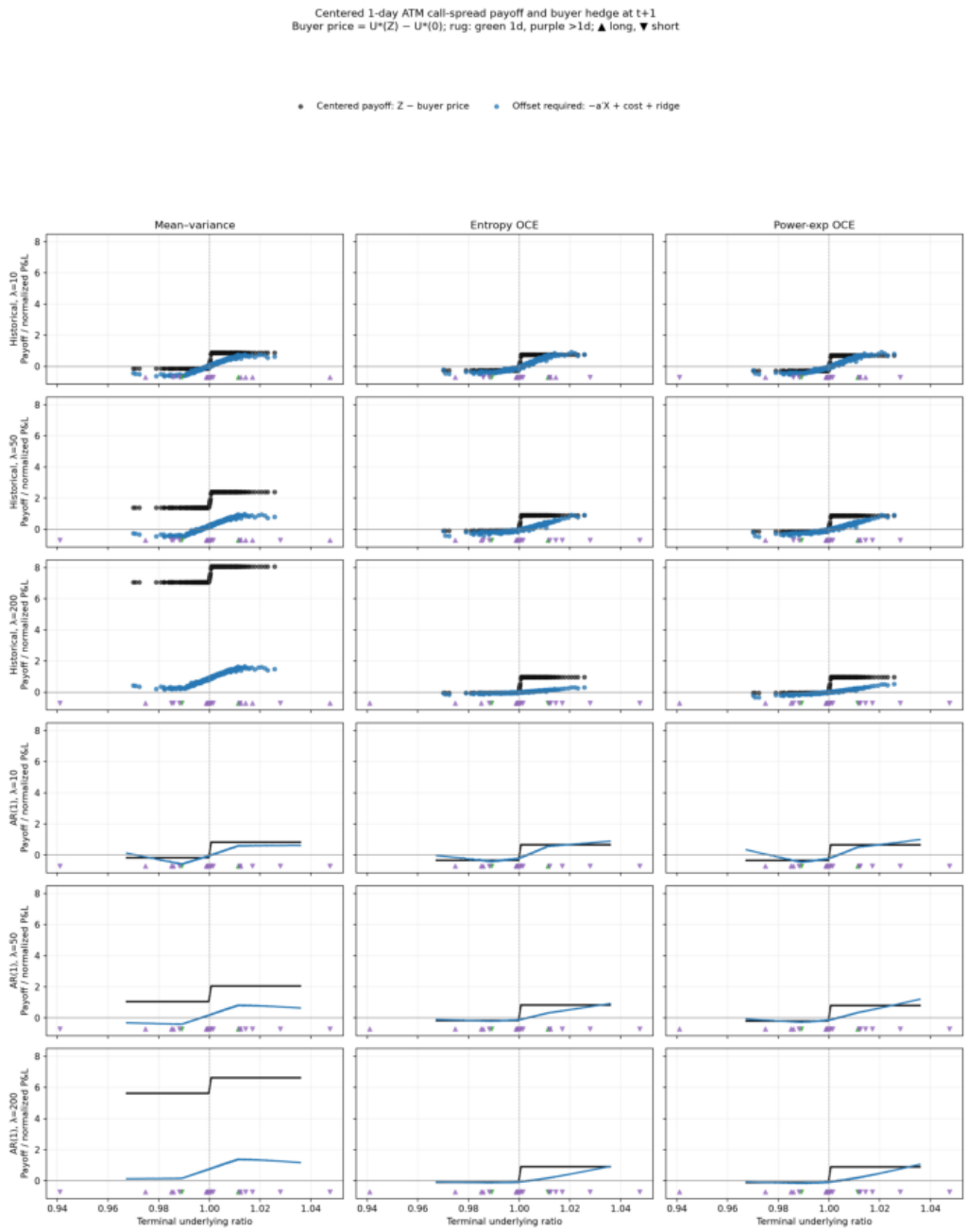
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



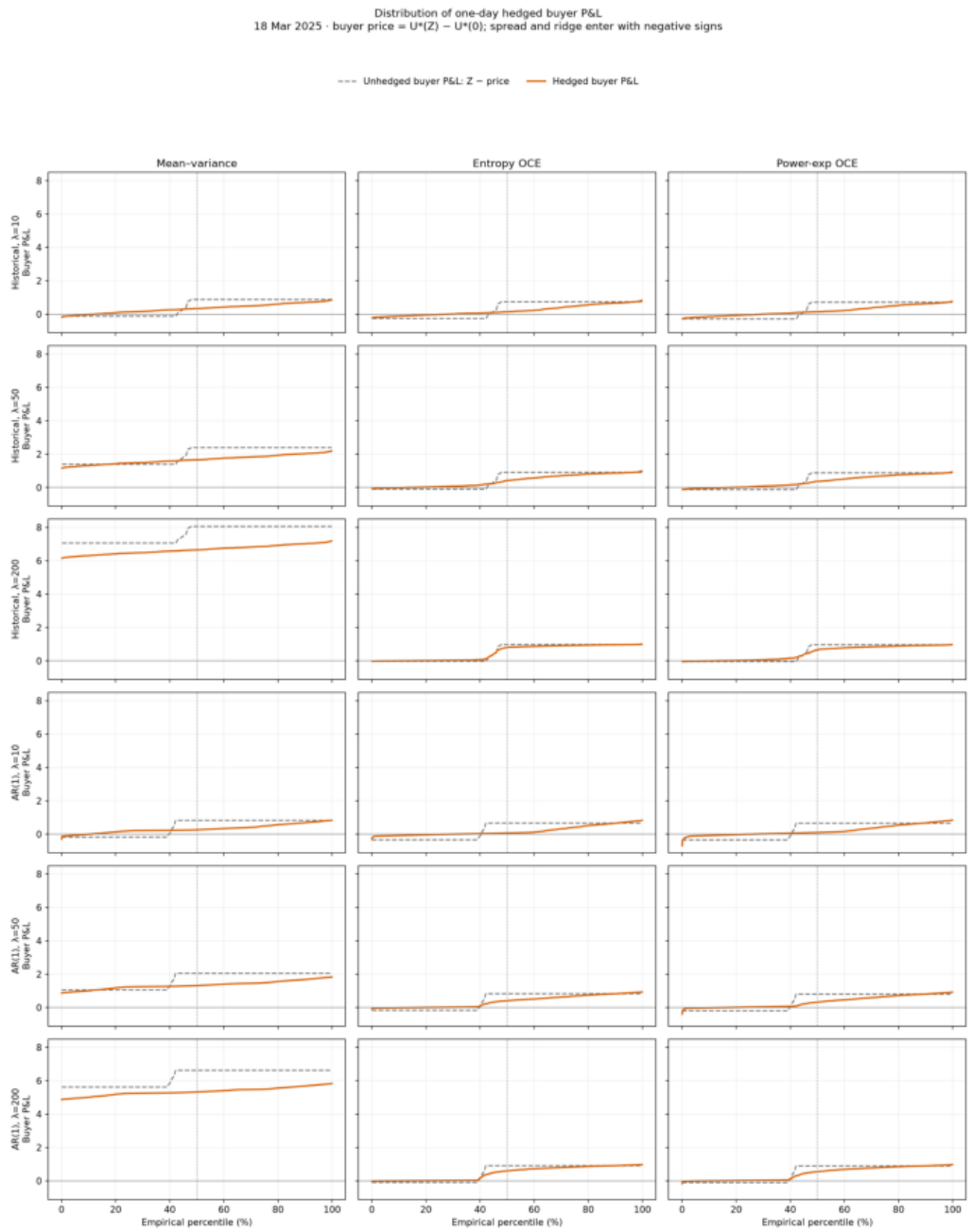
Buyer payoff and hedge

Buyer-side target payoff and replication under the original λ-scaled ridge specification.



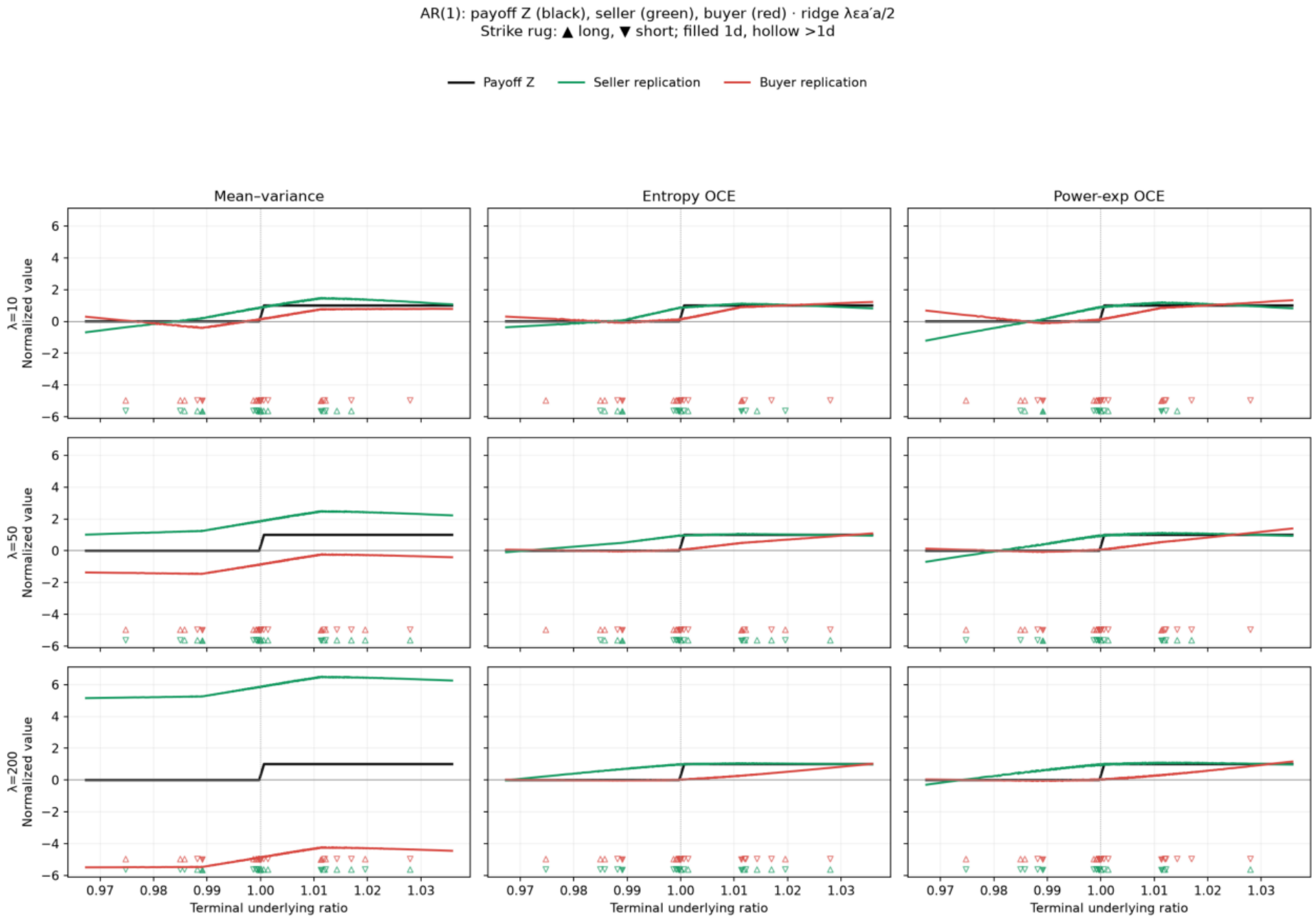
P&L quantiles — historical and AR(1)

Empirical residual-P&L percentiles for hedged and unhedged buyer/seller positions across utilities and risk aversions; λ-scaled ridge.



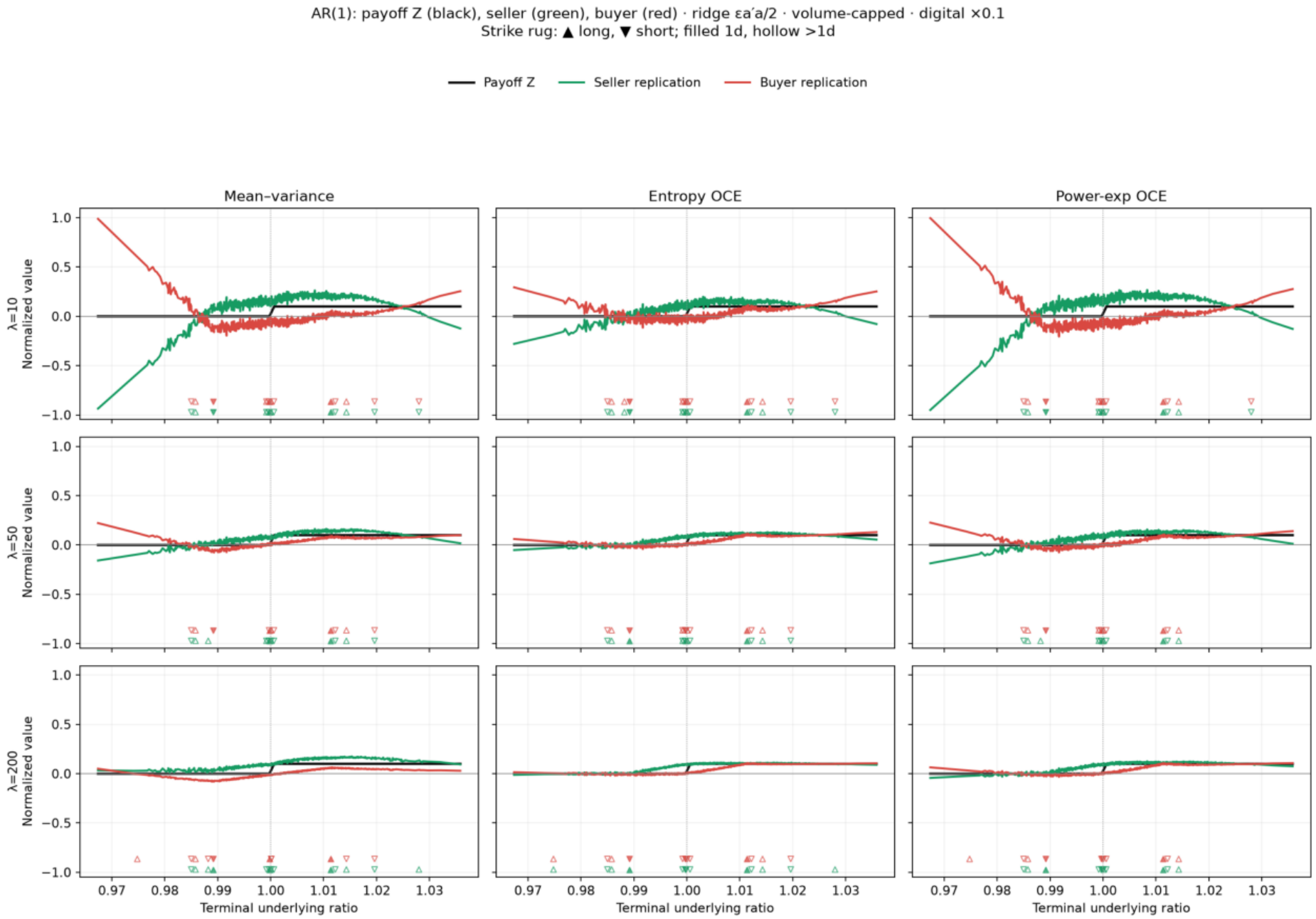
Payoff and hedge — AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; λ-scaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



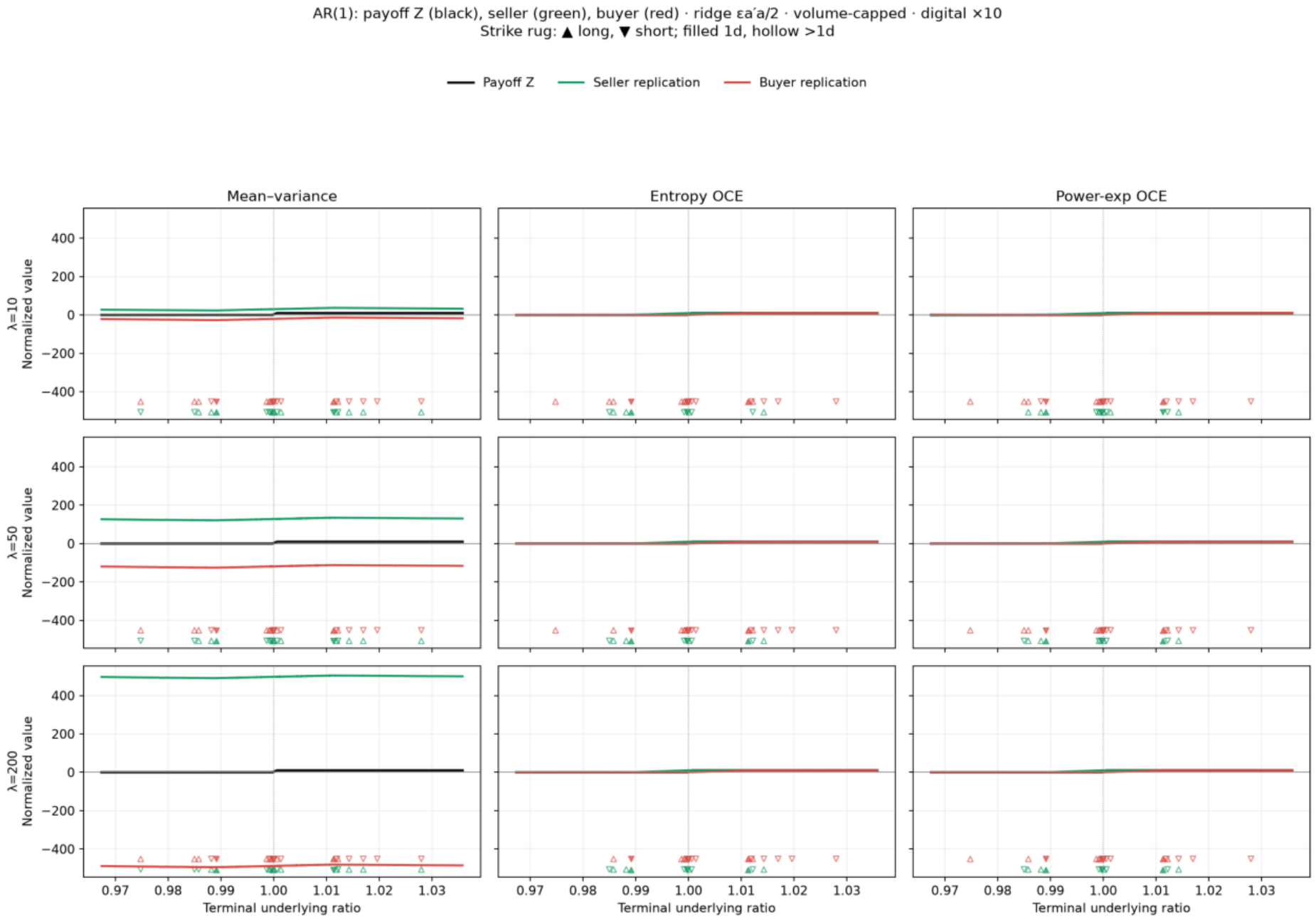
Payoff and hedge — AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 0.1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



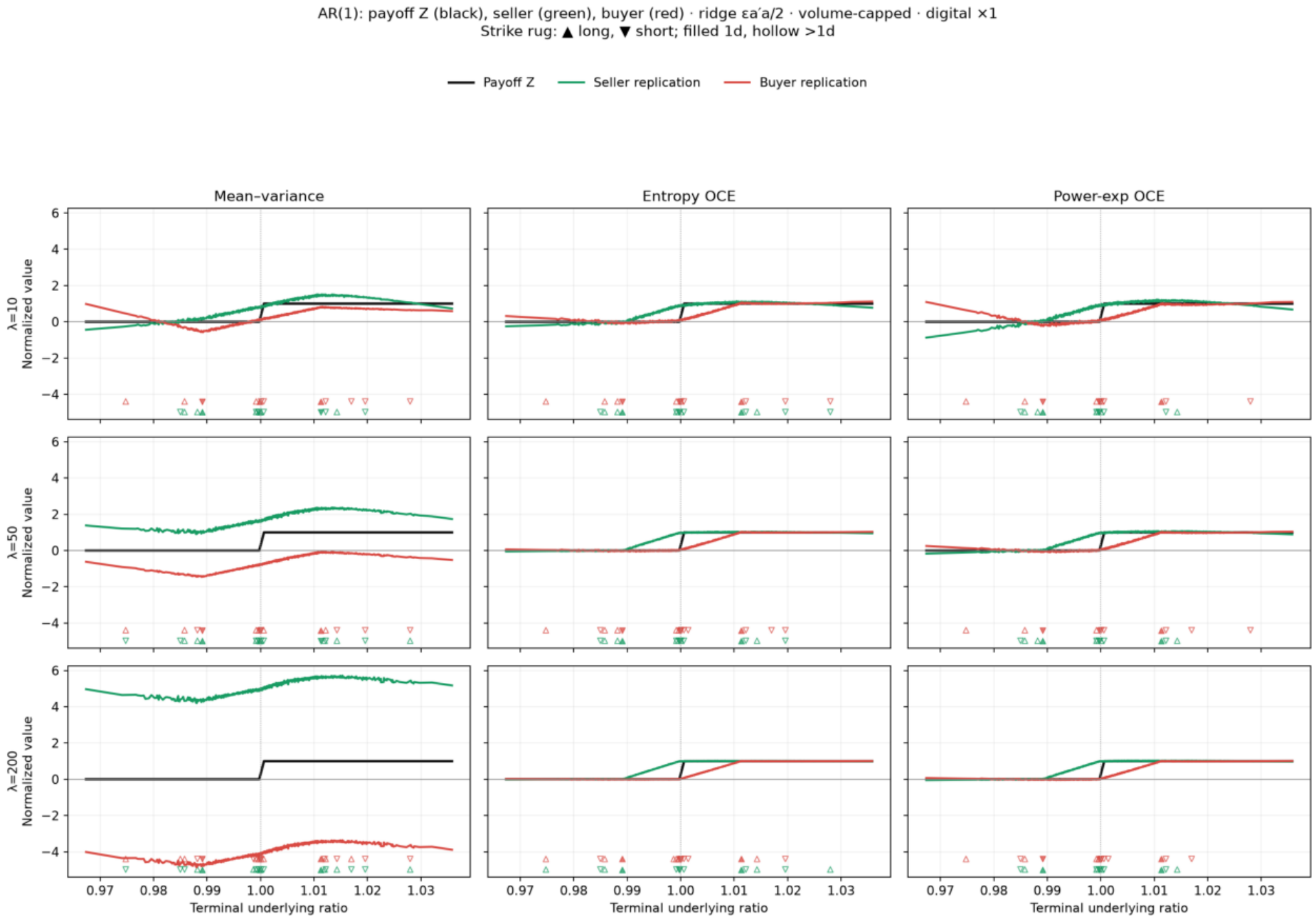
Payoff and hedge — AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



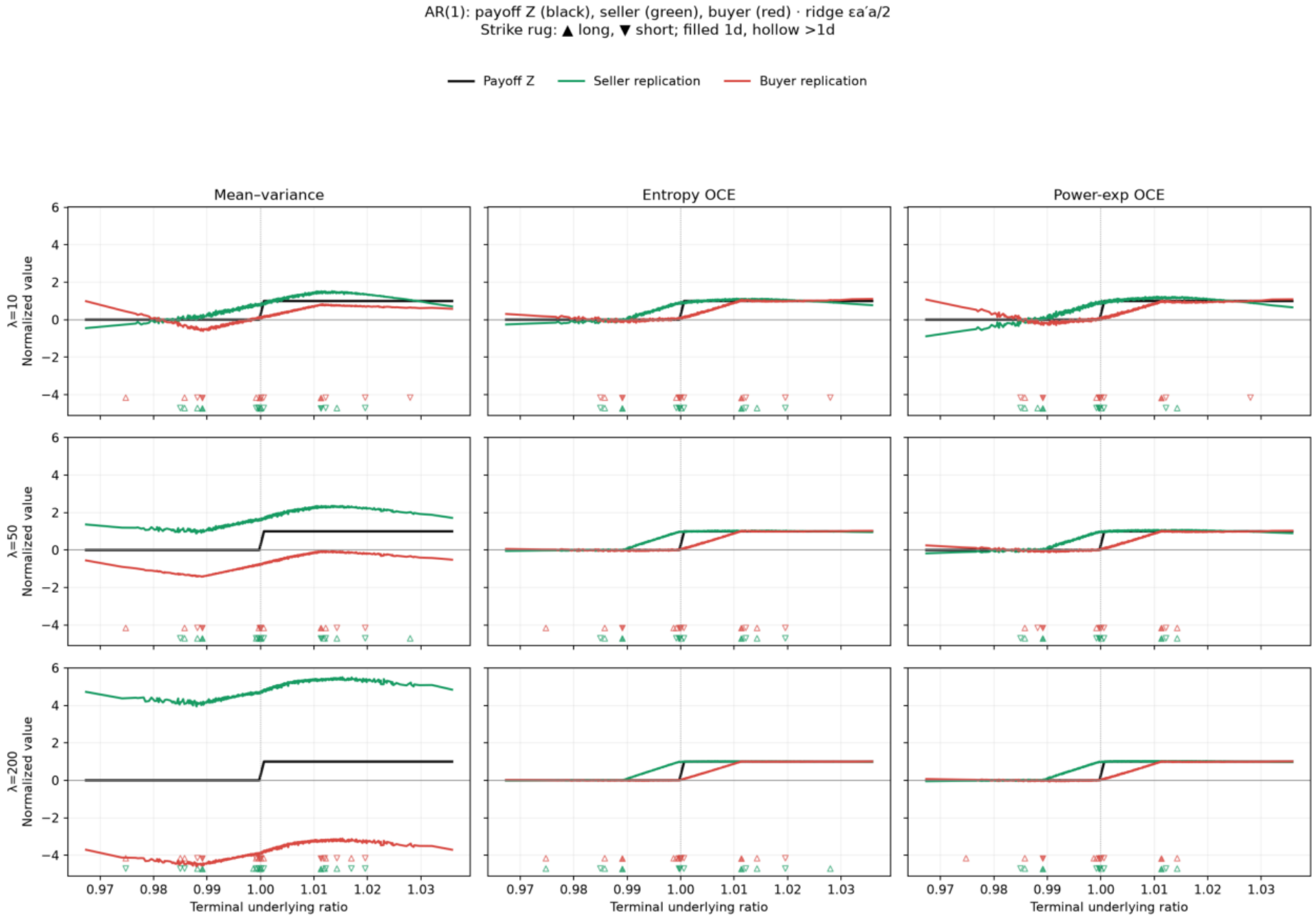
Payoff and hedge — AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



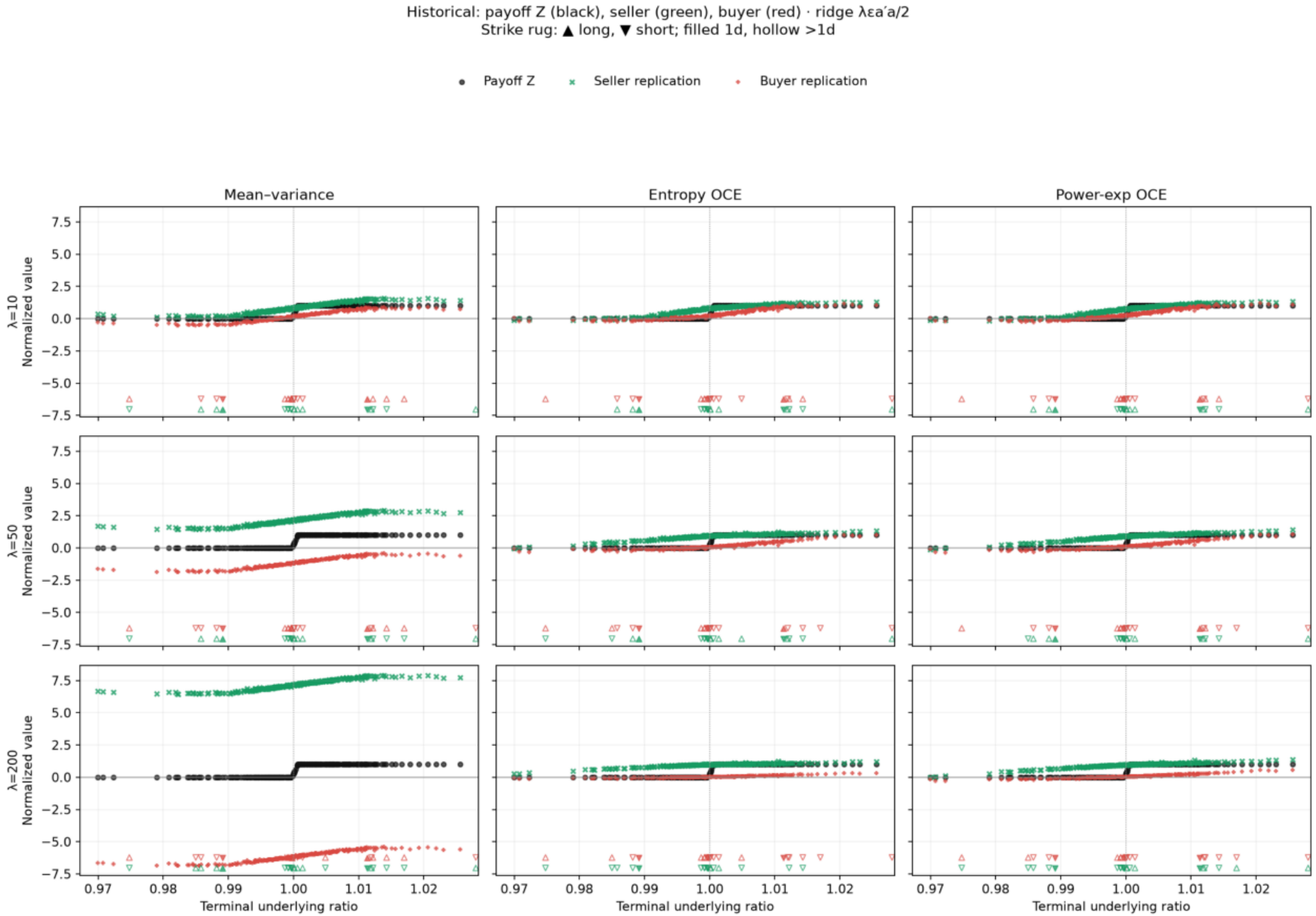
Payoff and hedge — AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



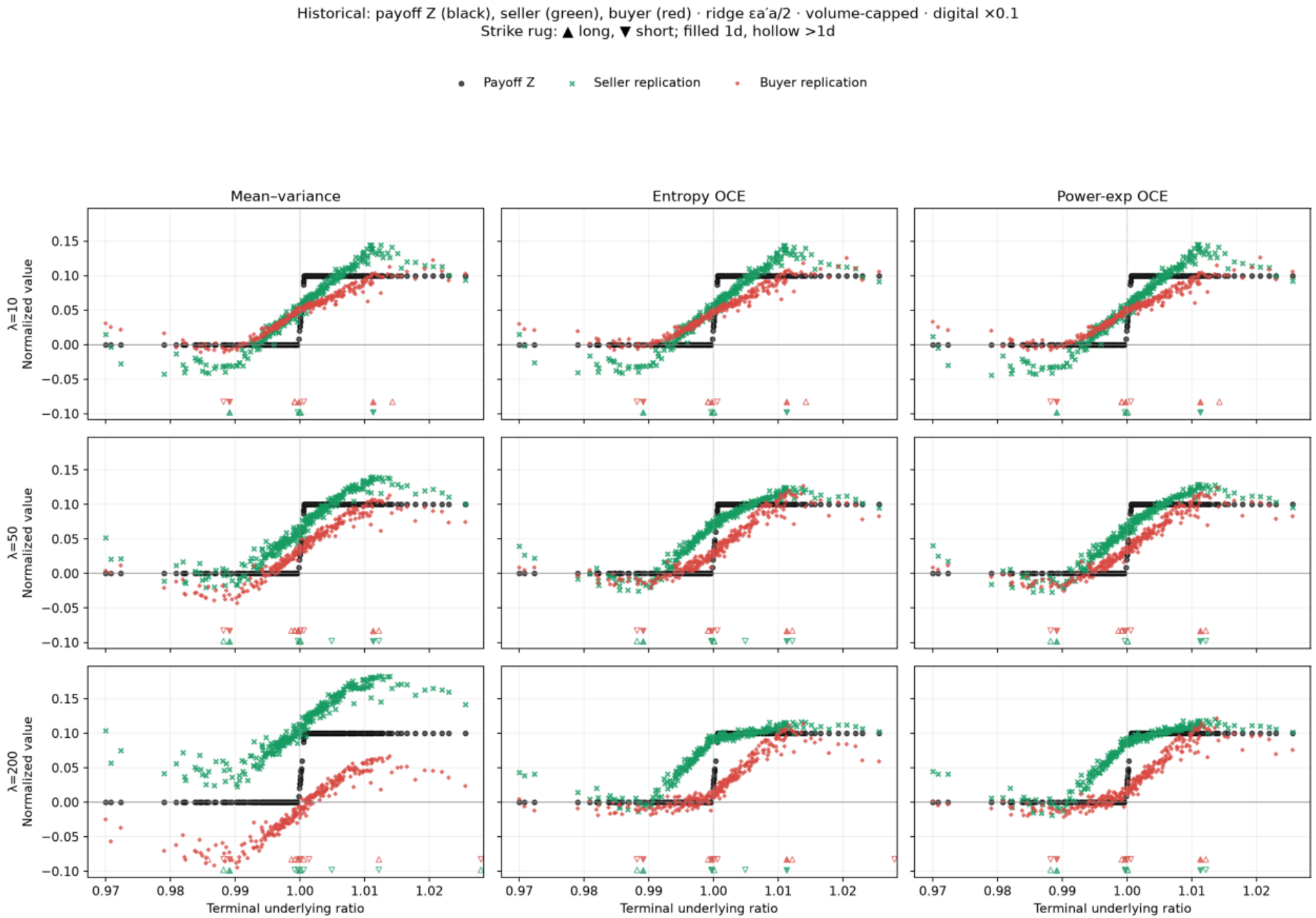
Payoff and hedge — historical

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; λ-scaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



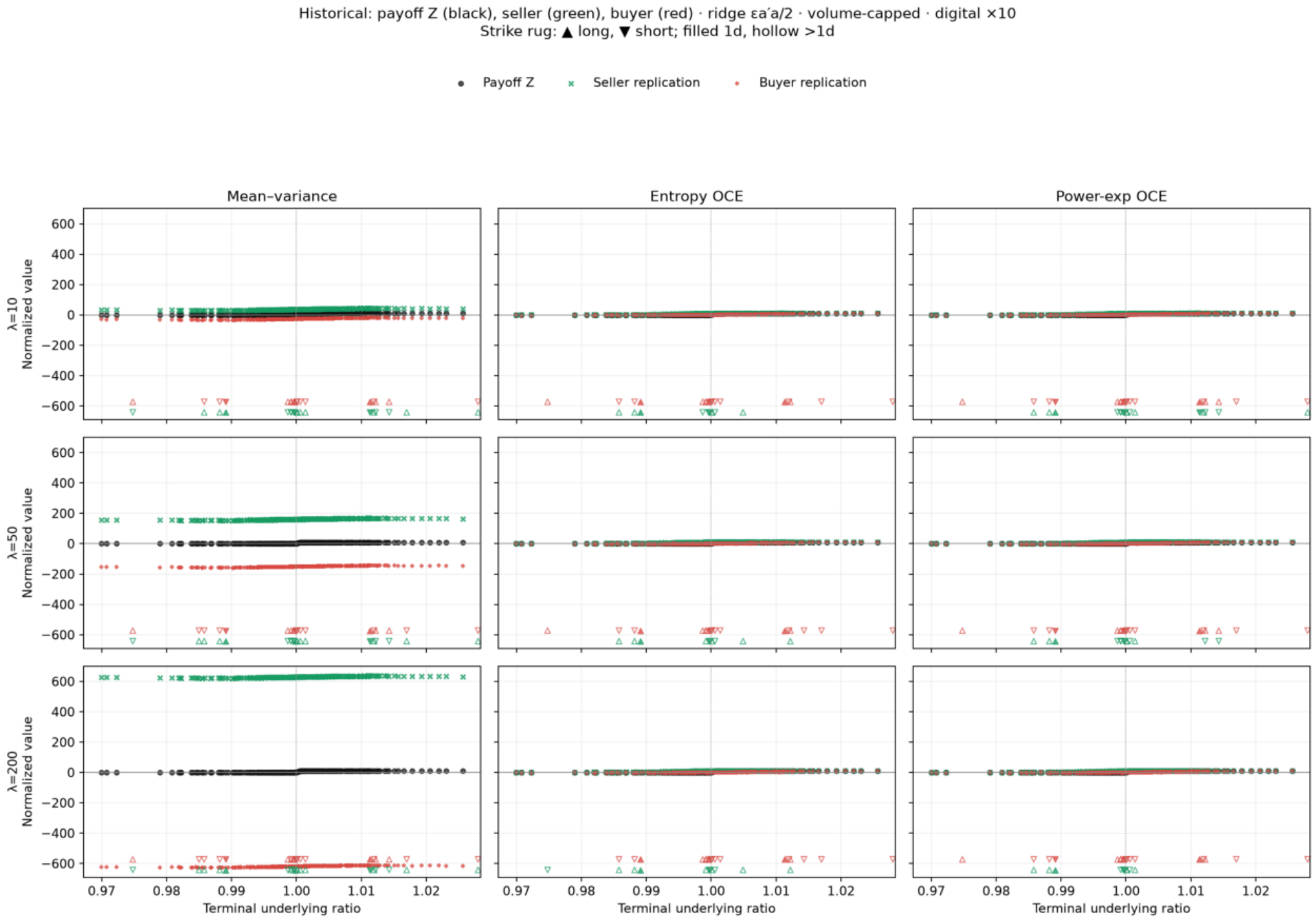
Payoff and hedge — historical

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 0.1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



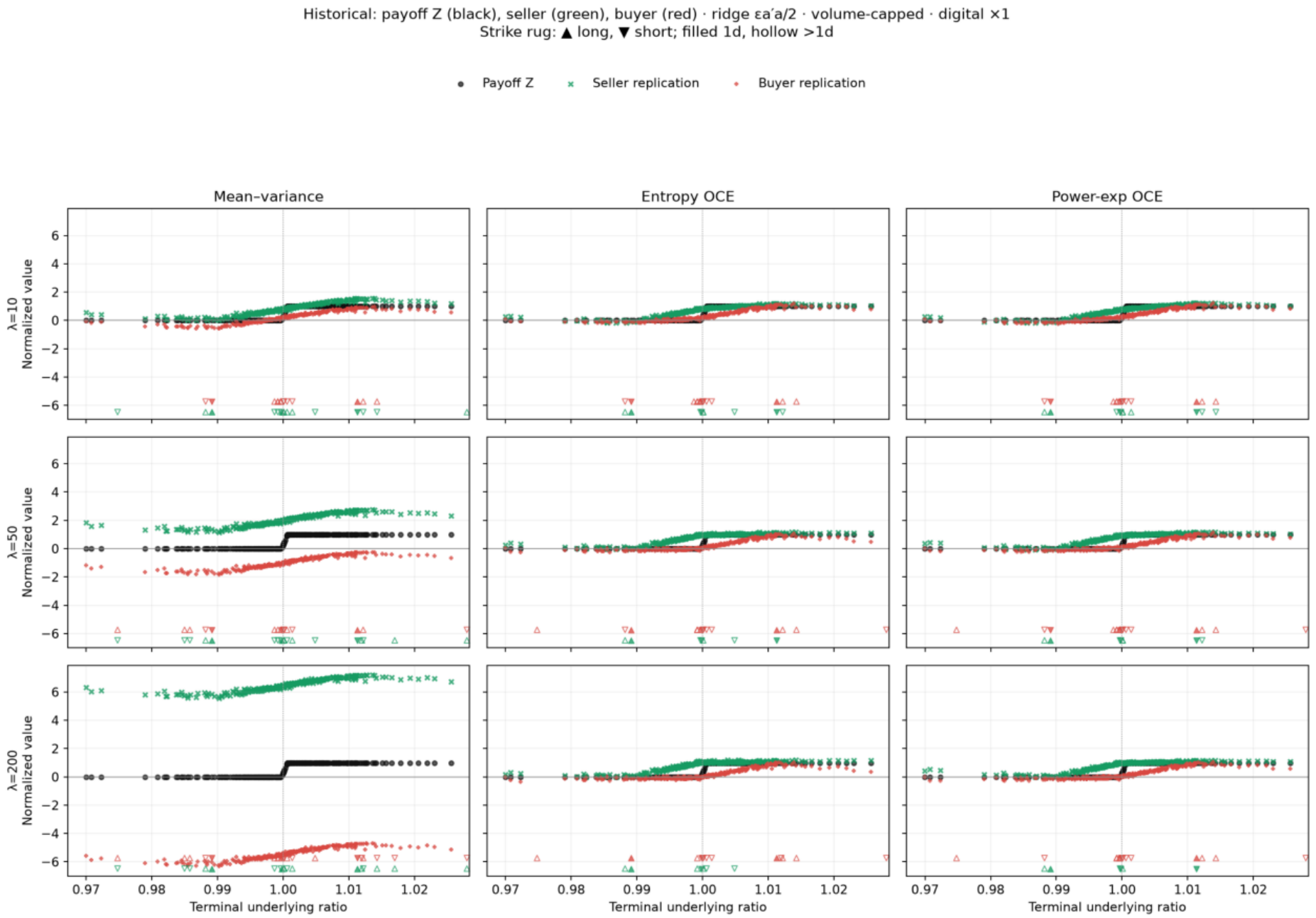
Payoff and hedge — historical

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



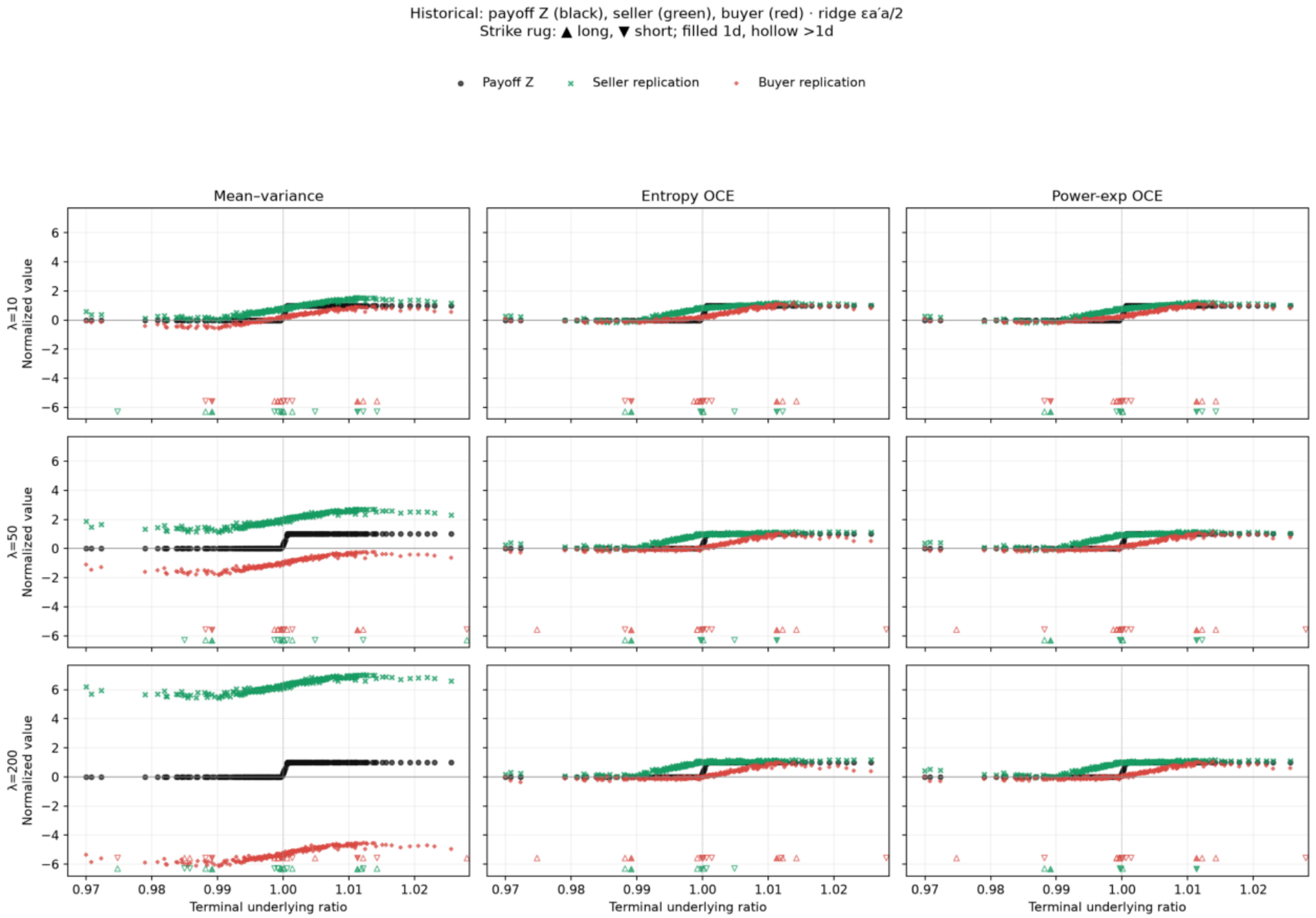
Payoff and hedge — historical

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



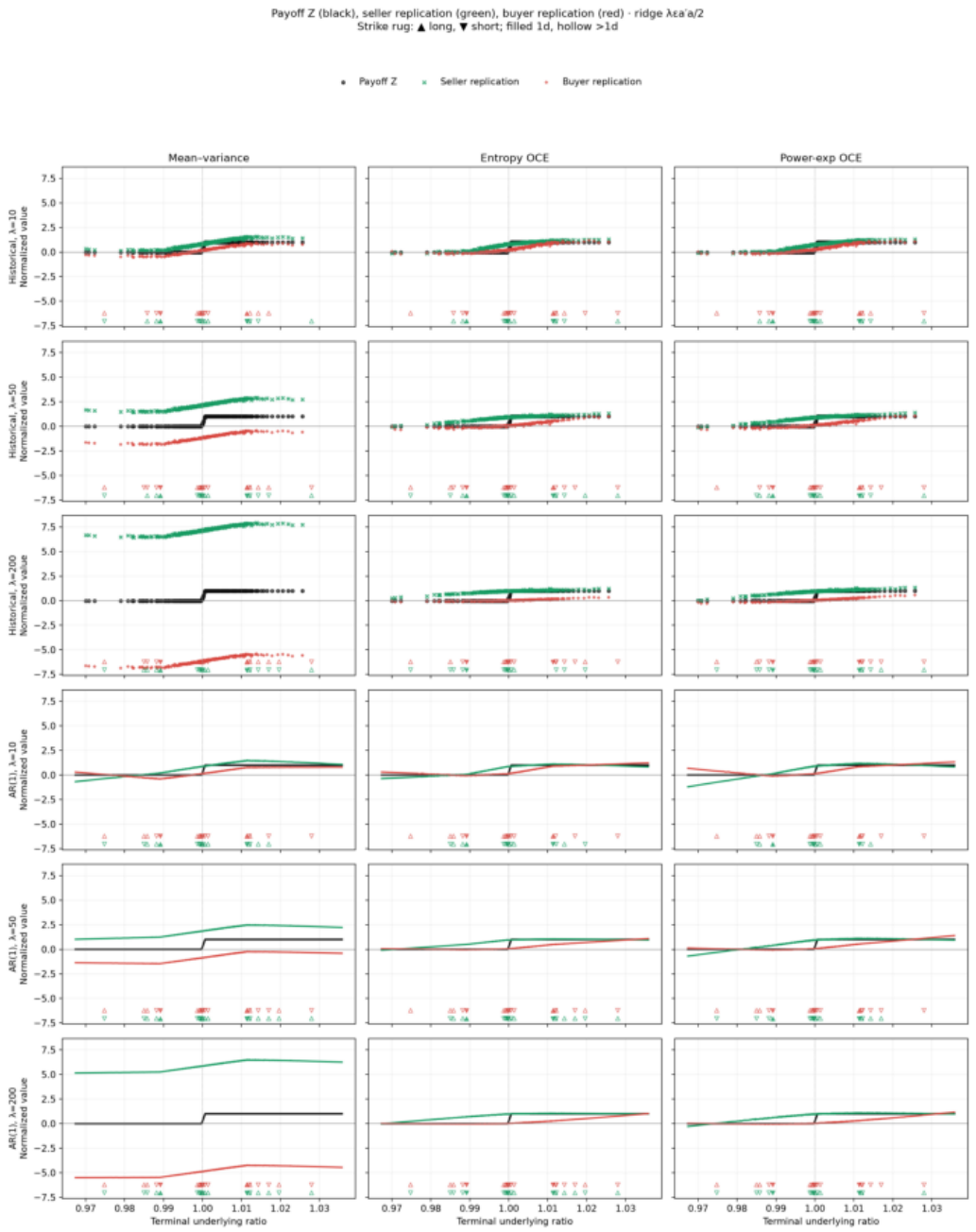
Payoff and hedge — historical

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



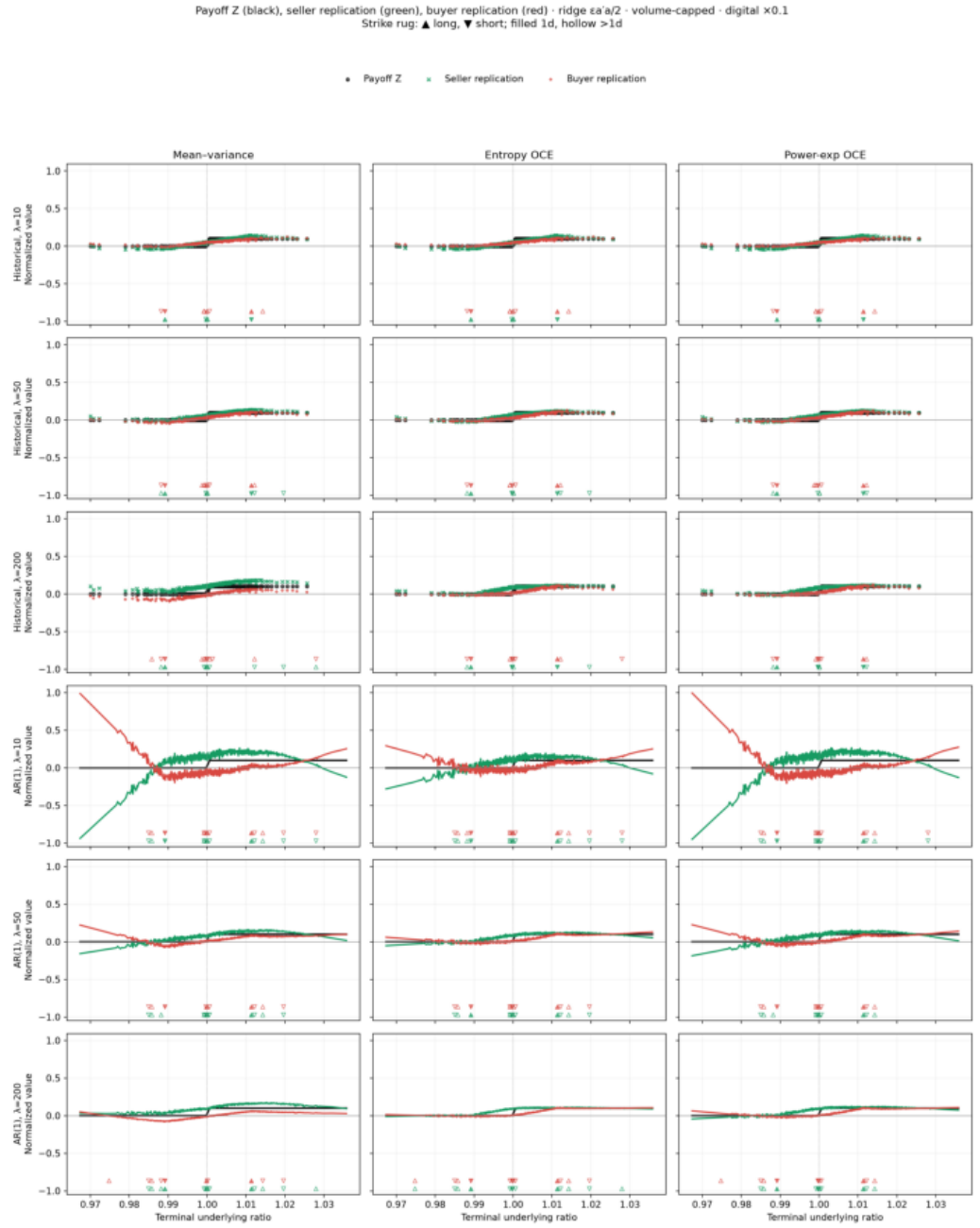
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; λ-scaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



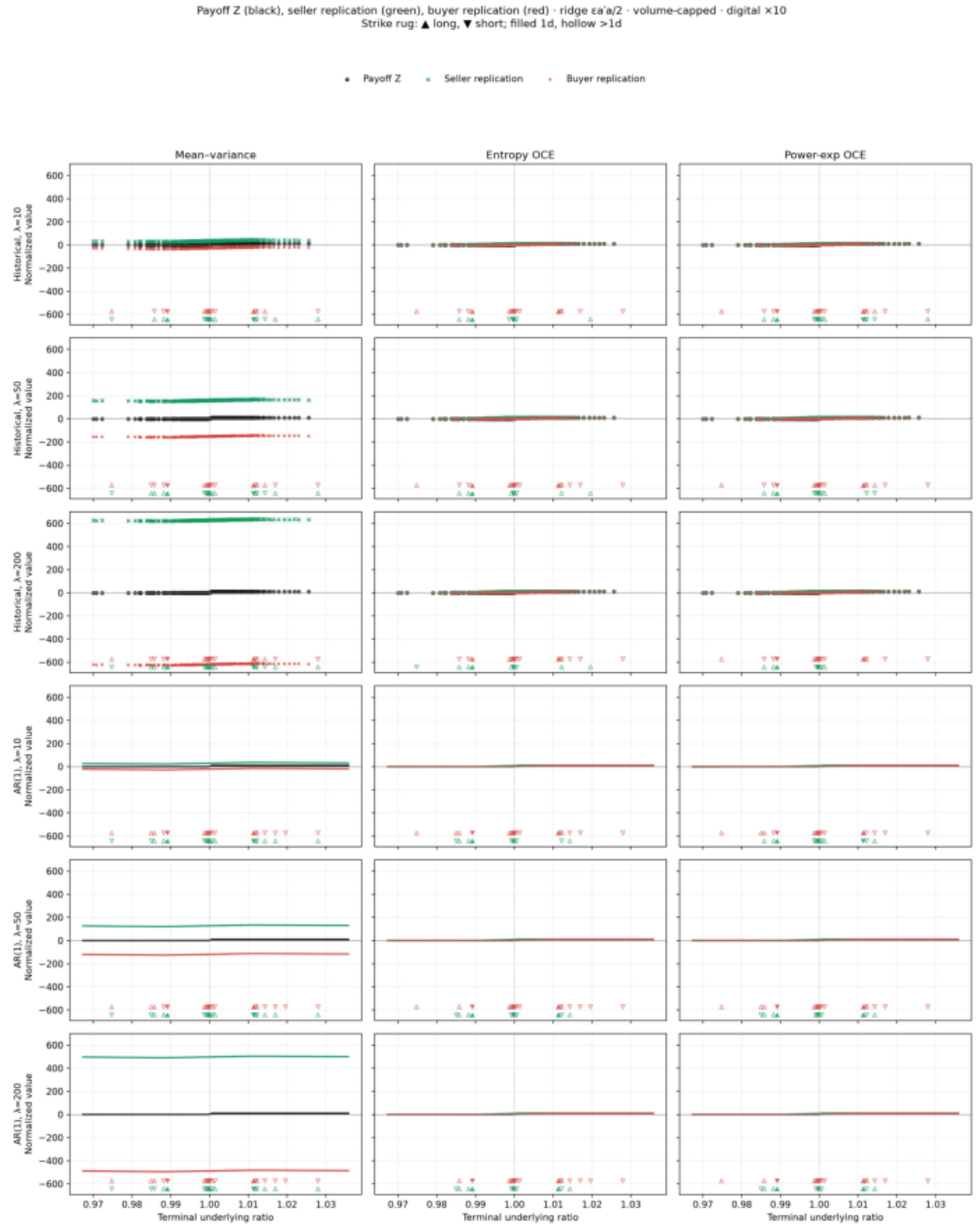
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 0.1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



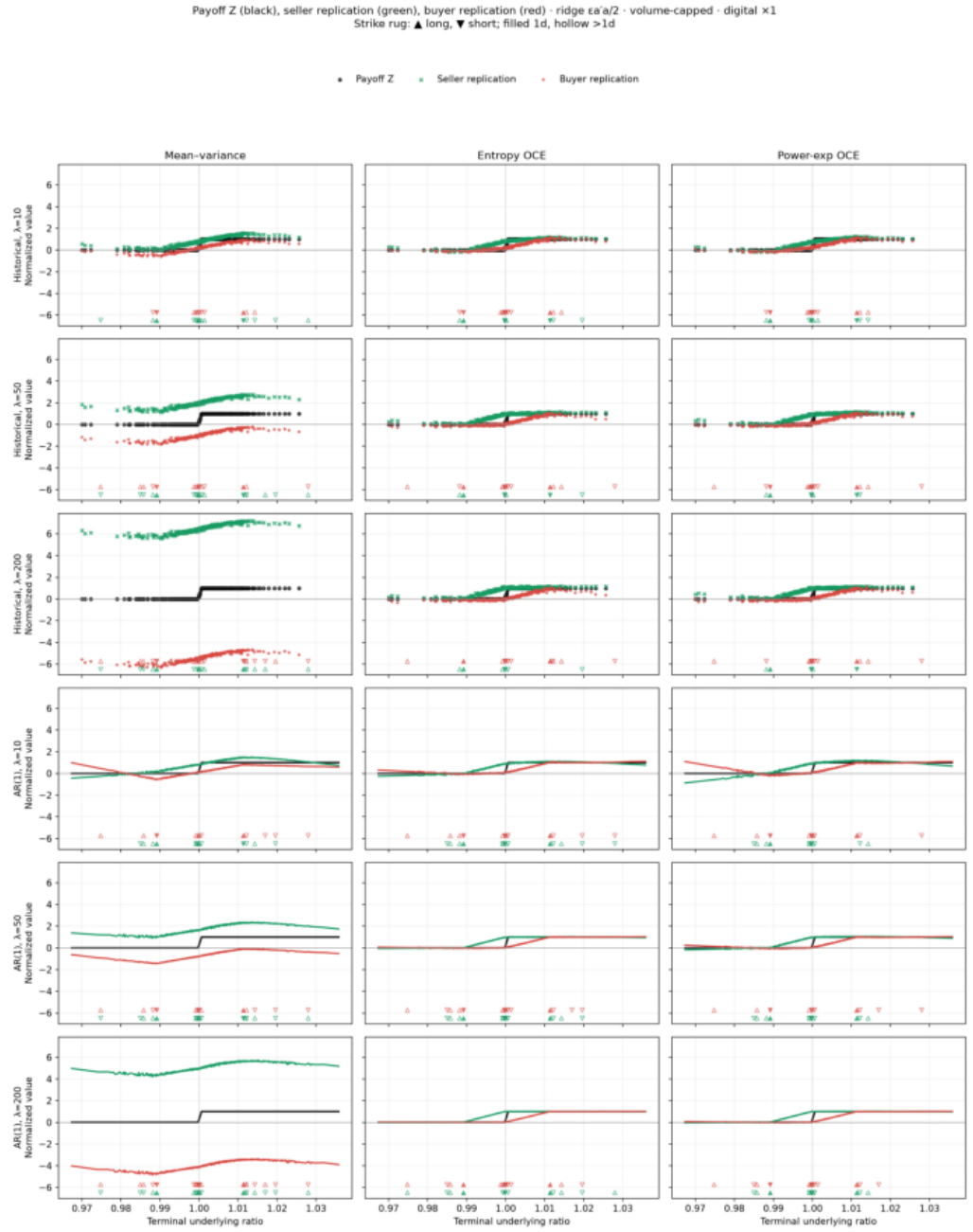
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



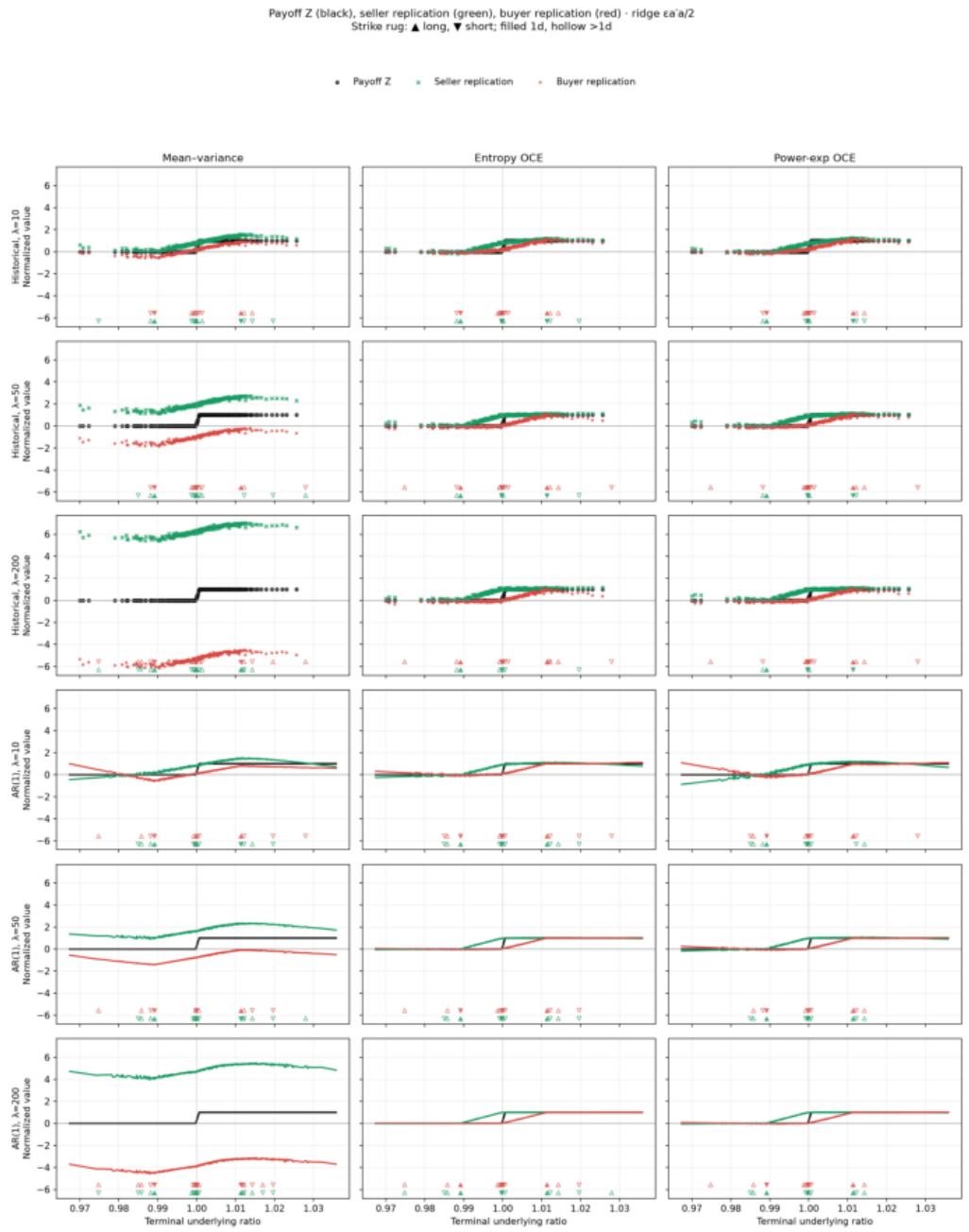
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



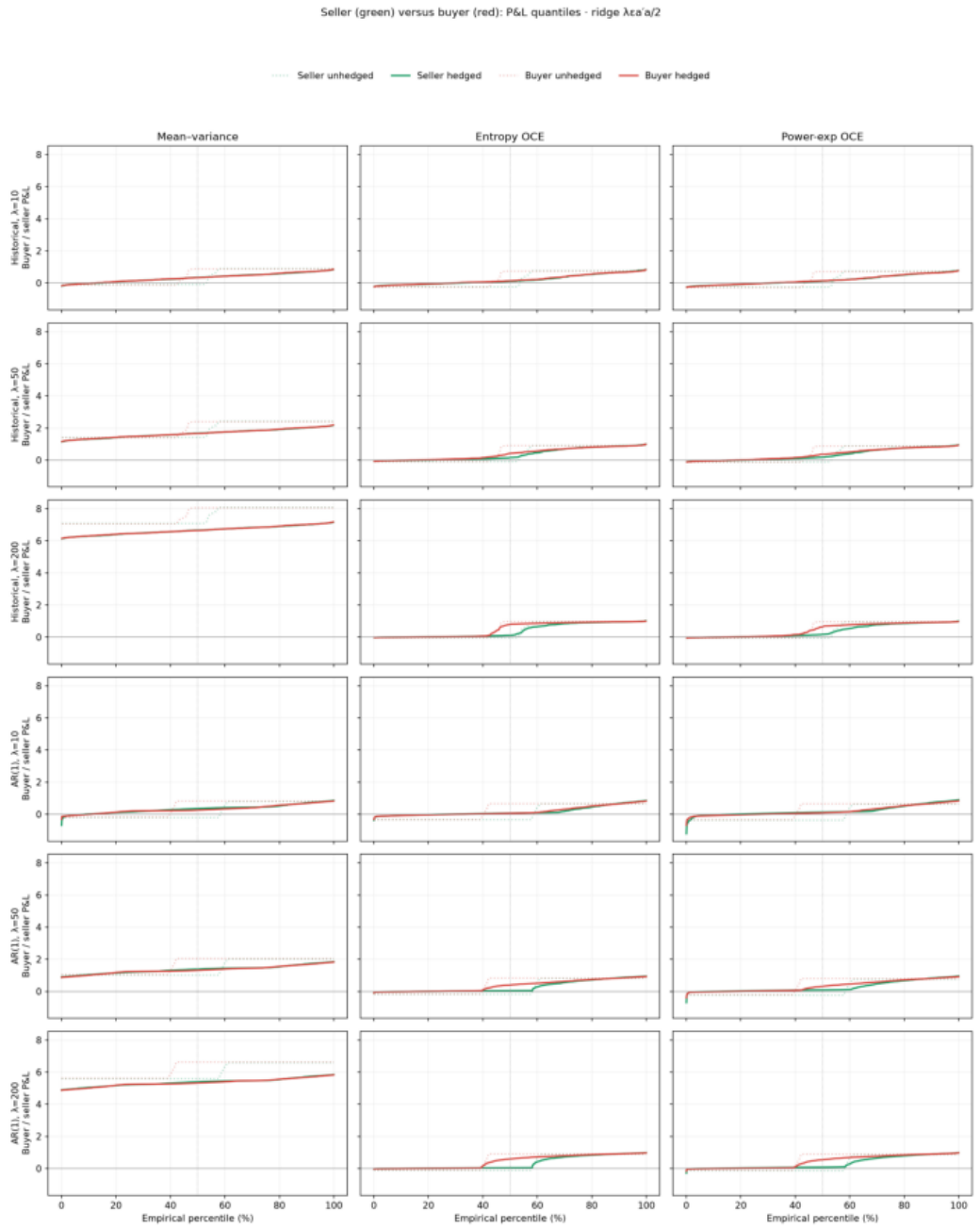
Payoff and hedge — historical and AR(1)

Target payoff in black, seller replication in green and buyer replication in red across utilities and risk aversions; classic unscaled ridge. Strike markers identify traded relative strikes; ▲ is long and ▼ is short.



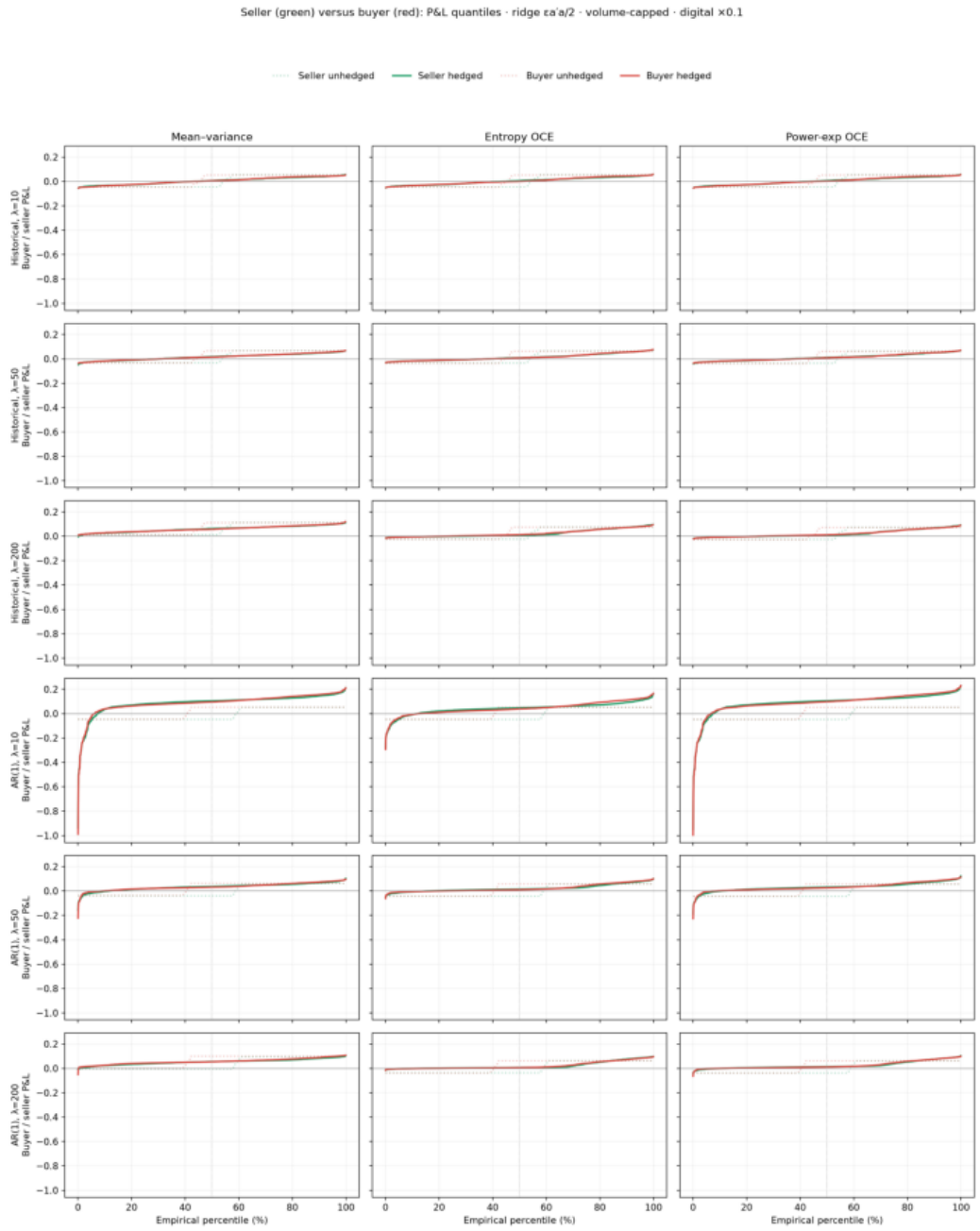
P&L quantiles — historical and AR(1)

Empirical residual-P&L percentiles for hedged and unhedged buyer/seller positions across utilities and risk aversions; λ-scaled ridge.



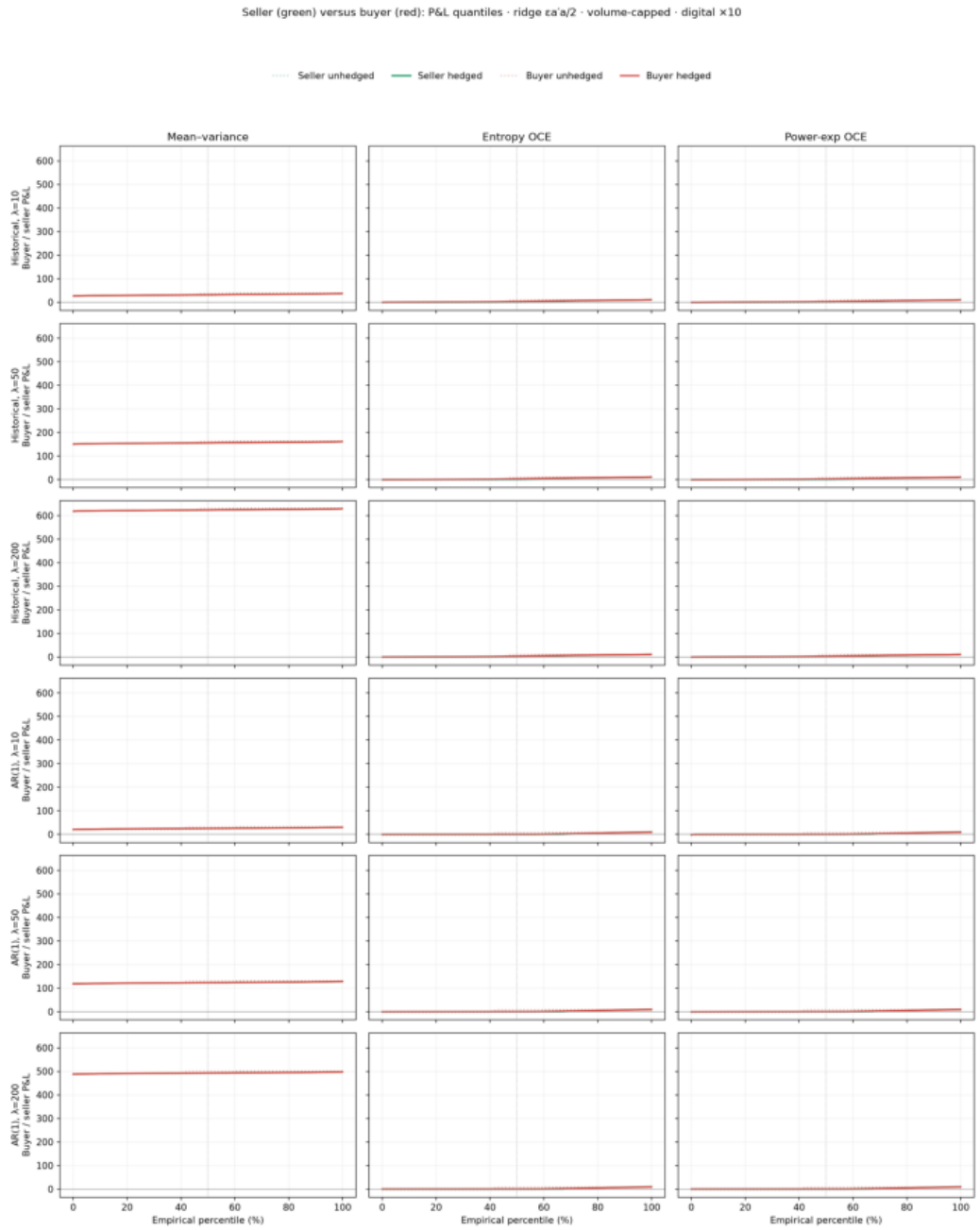
P&L quantiles — historical and AR(1)

Empirical residual-P&L percentiles for hedged and unhedged buyer/seller positions across utilities and risk aversions; classic unscaled ridge. Digital notional is 0.1×, with each position hard-capped by reported target-day volume.



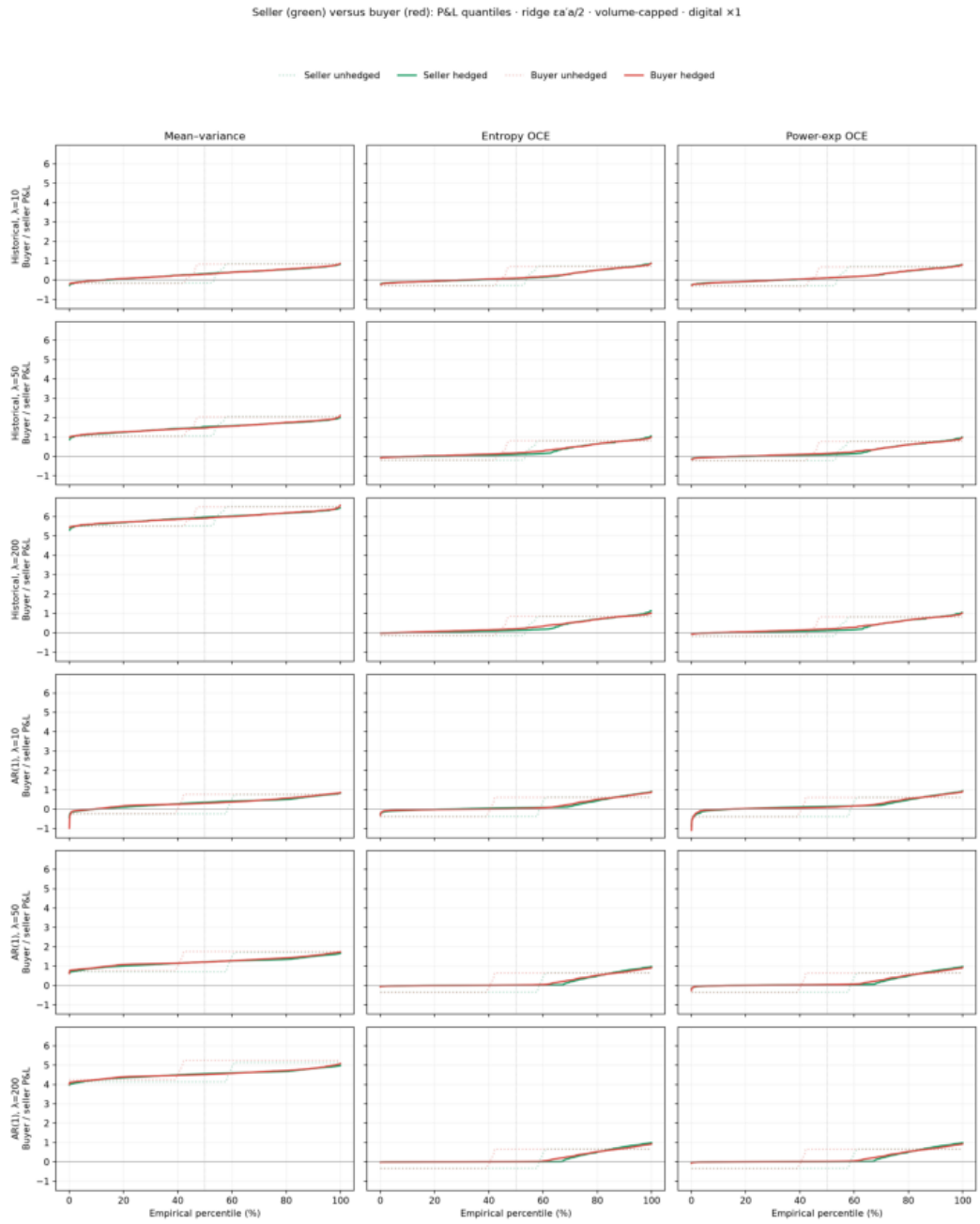
P&L quantiles — historical and AR(1)

Empirical residual-P&L percentiles for hedged and unhedged buyer/seller positions across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume.



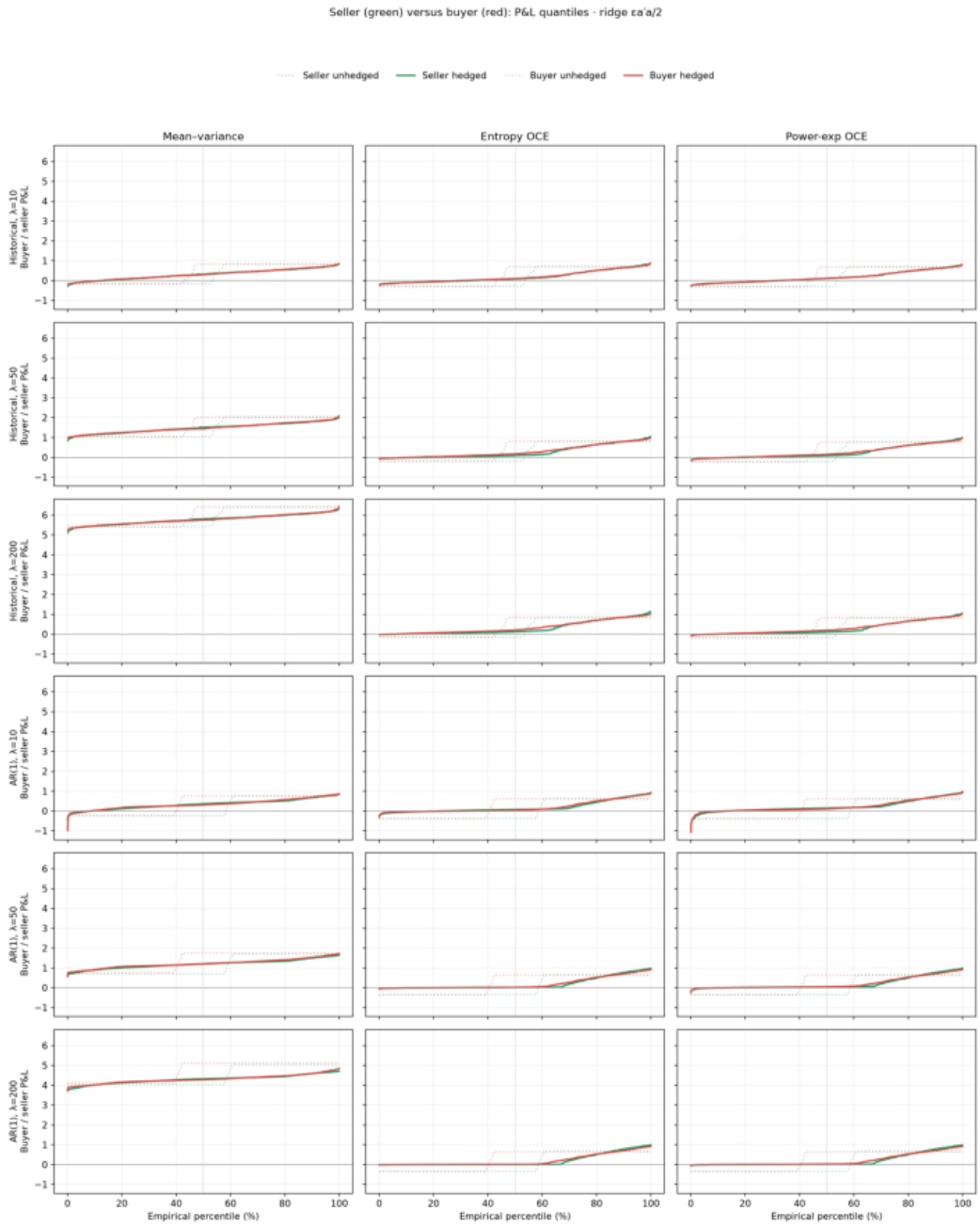
P&L quantiles — historical and AR(1)

Empirical residual-P&L percentiles for hedged and unhedged buyer/seller positions across utilities and risk aversions; classic unscaled ridge. Digital notional is 1×, with each position hard-capped by reported target-day volume.



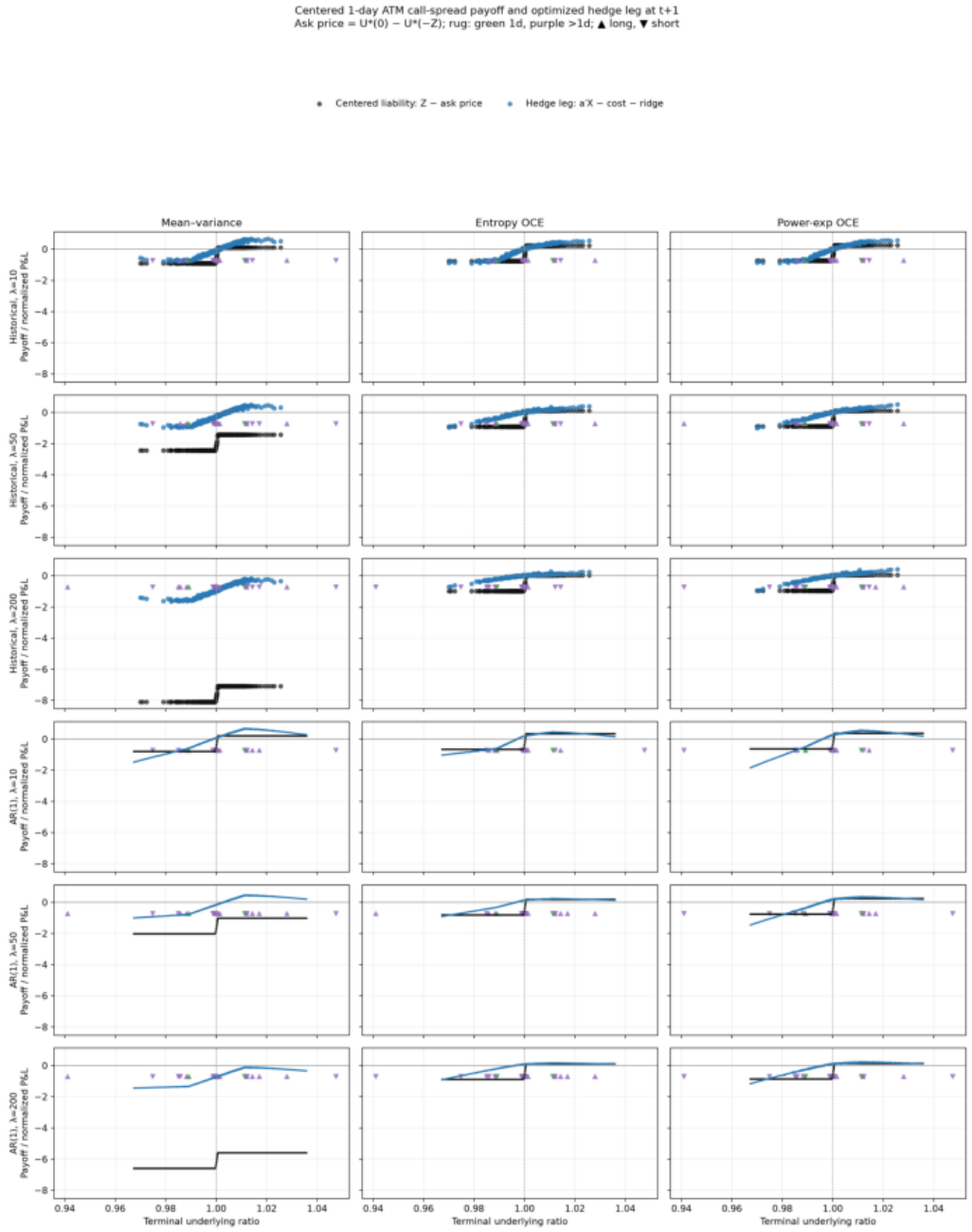
P&L quantiles — historical and AR(1)

Empirical residual-P&L percentiles for hedged and unhedged buyer/seller positions across utilities and risk aversions; classic unscaled ridge.



Baseline payoff and hedge

Baseline seller experiment showing the target payoff and hedge value as risk aversion changes.



P&L quantiles — historical and AR(1)

Empirical residual-P&L percentiles for hedged and unhedged buyer/seller positions across utilities and risk aversions; classic unscaled ridge.

